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FORD SMART MOBILITY



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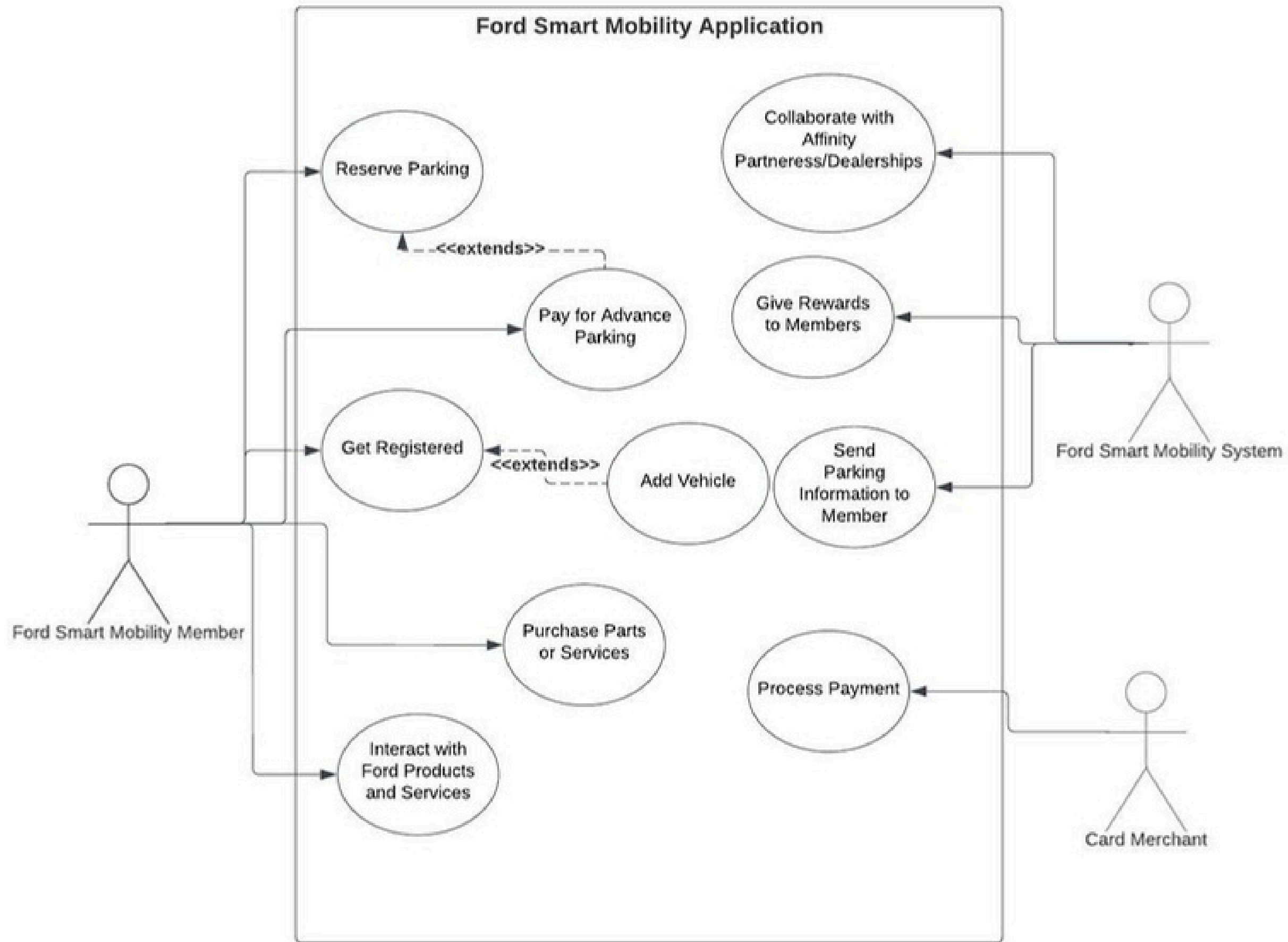


**Loyalty
Program**

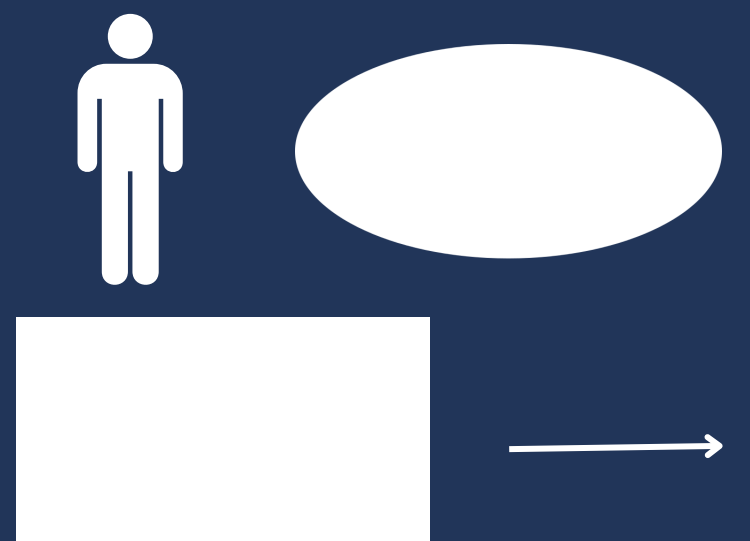
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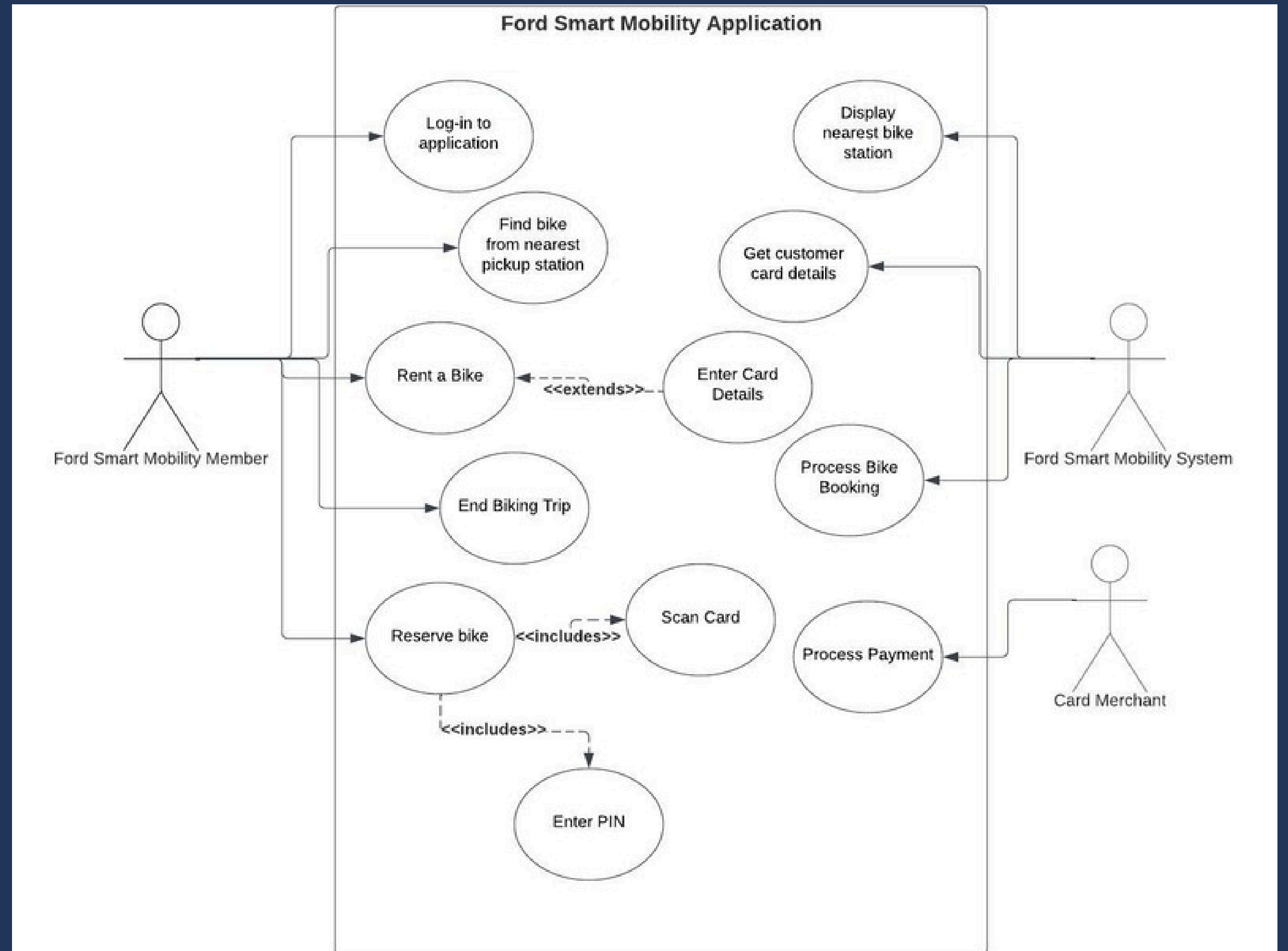
BEHAVIOURAL DIAGRAMS



LOYALTY SCHEME USE CASE

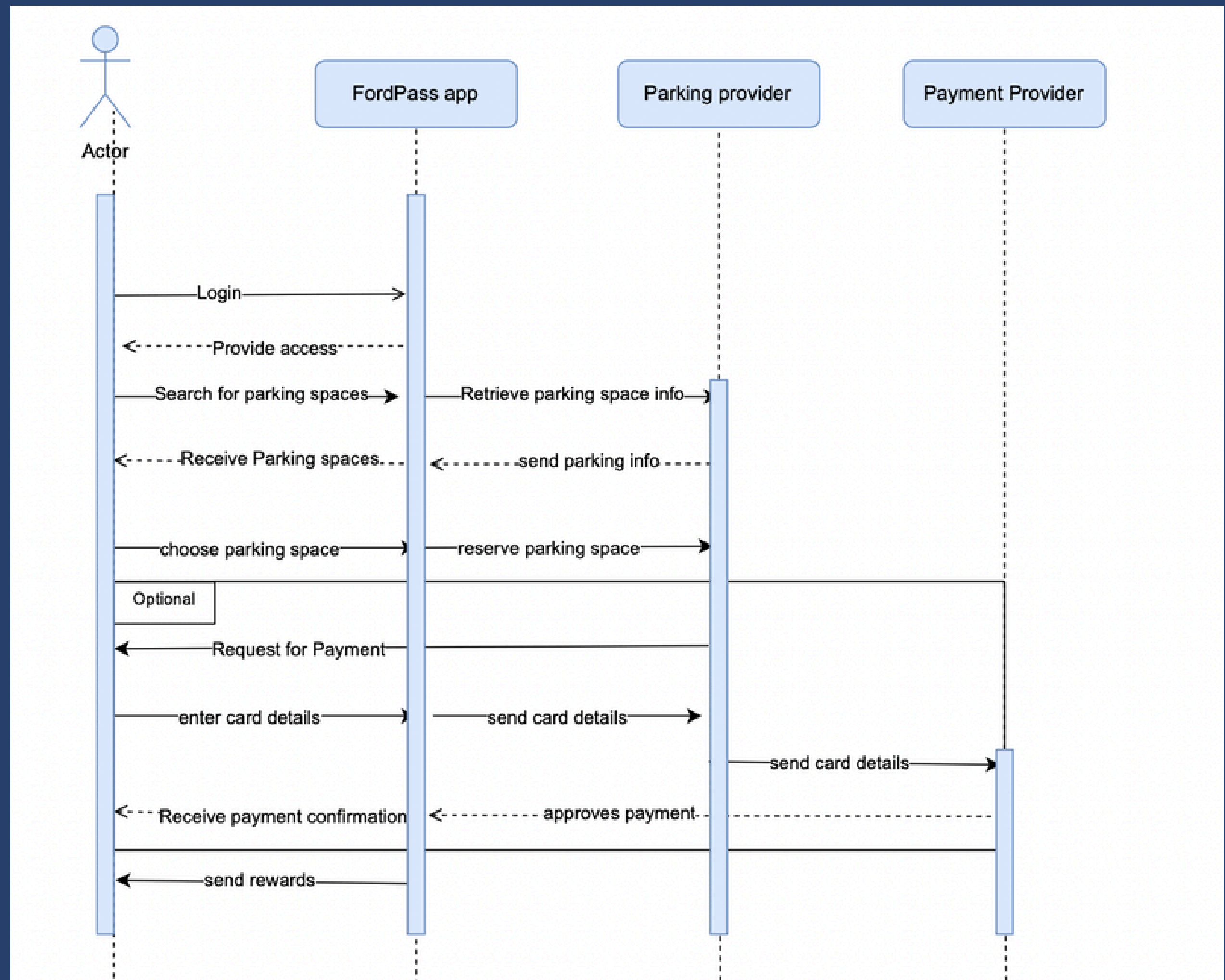


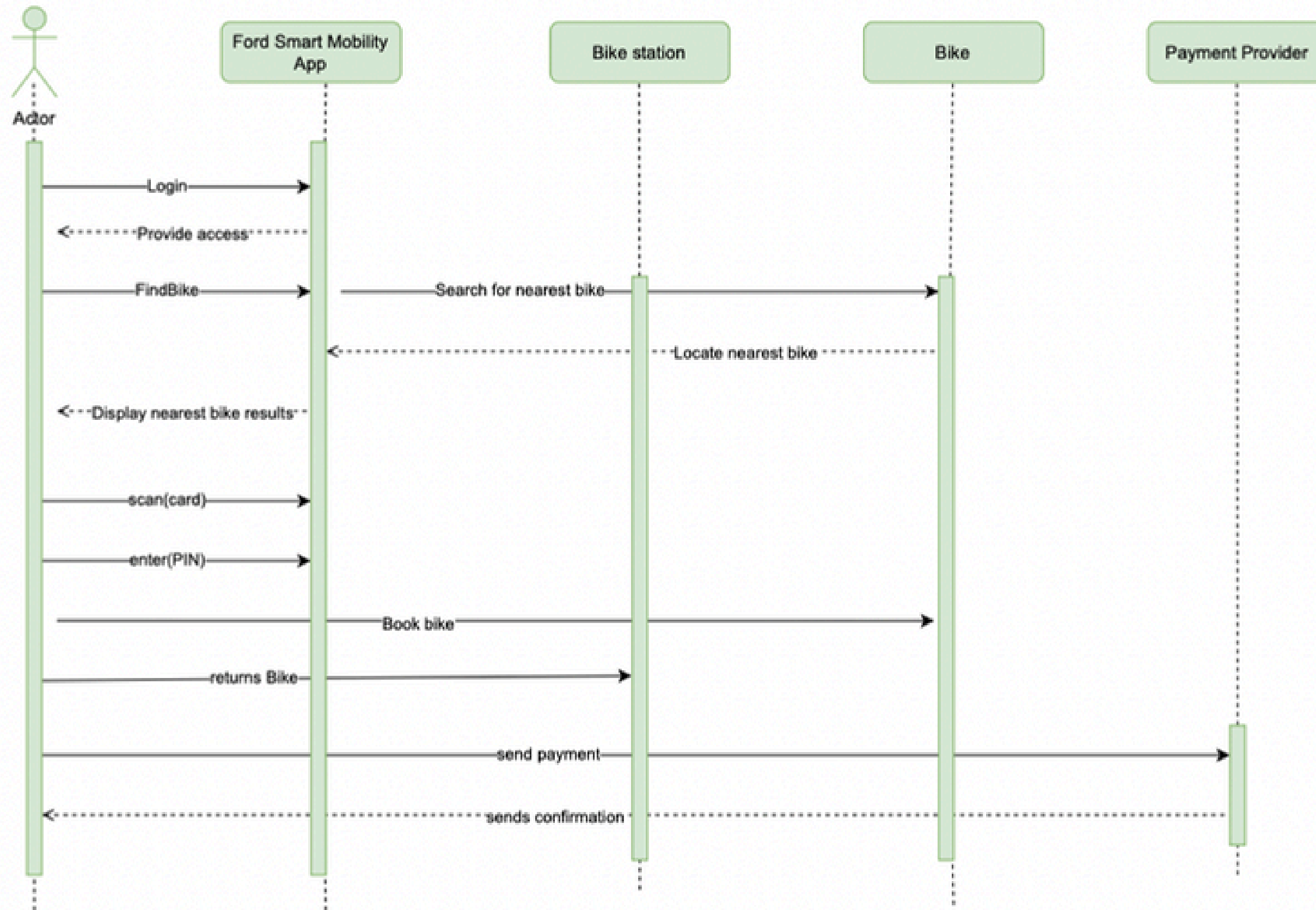
BIKE SHARE USE CASE





PARKING SEQUENCE DIAGRAM

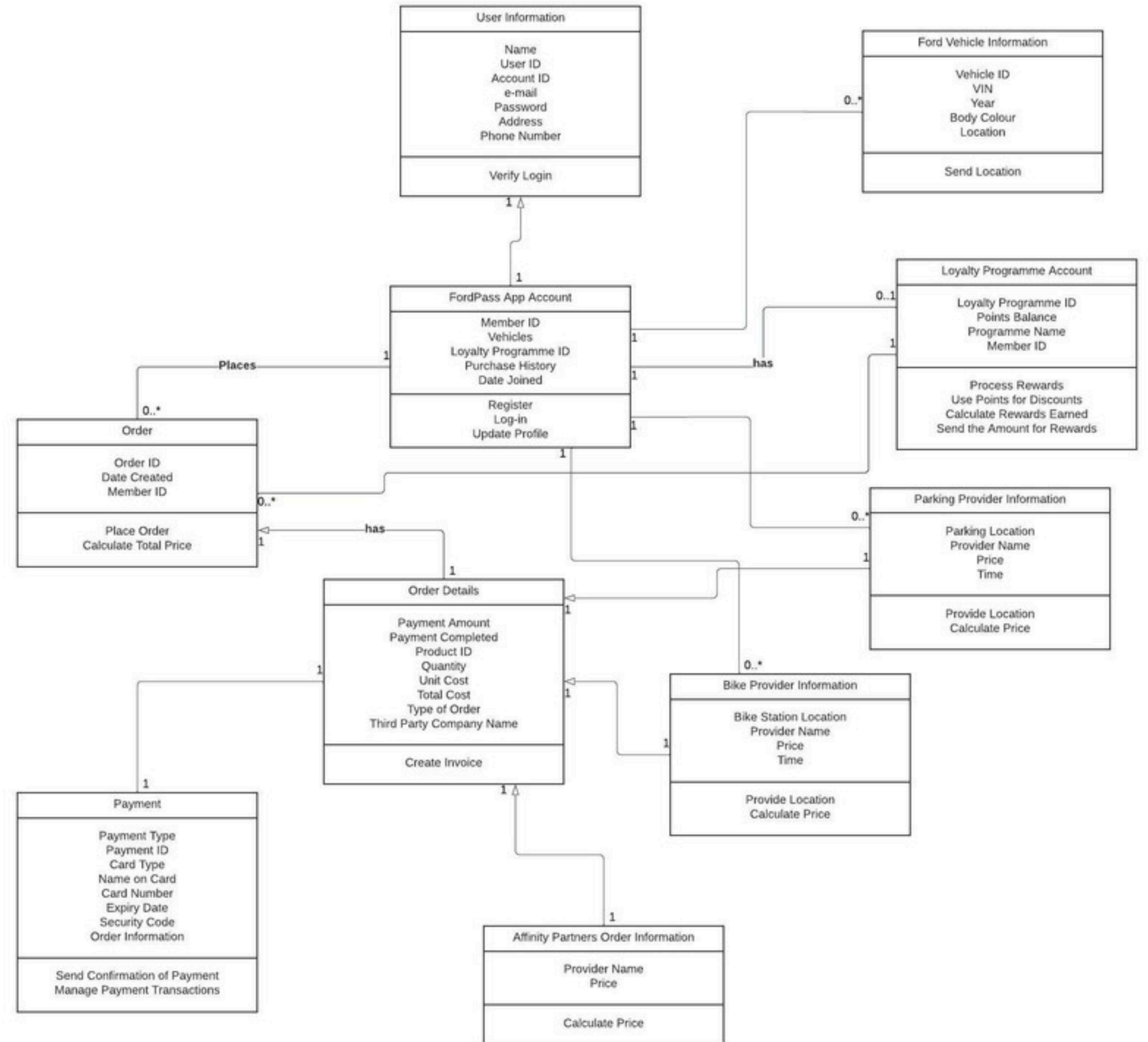




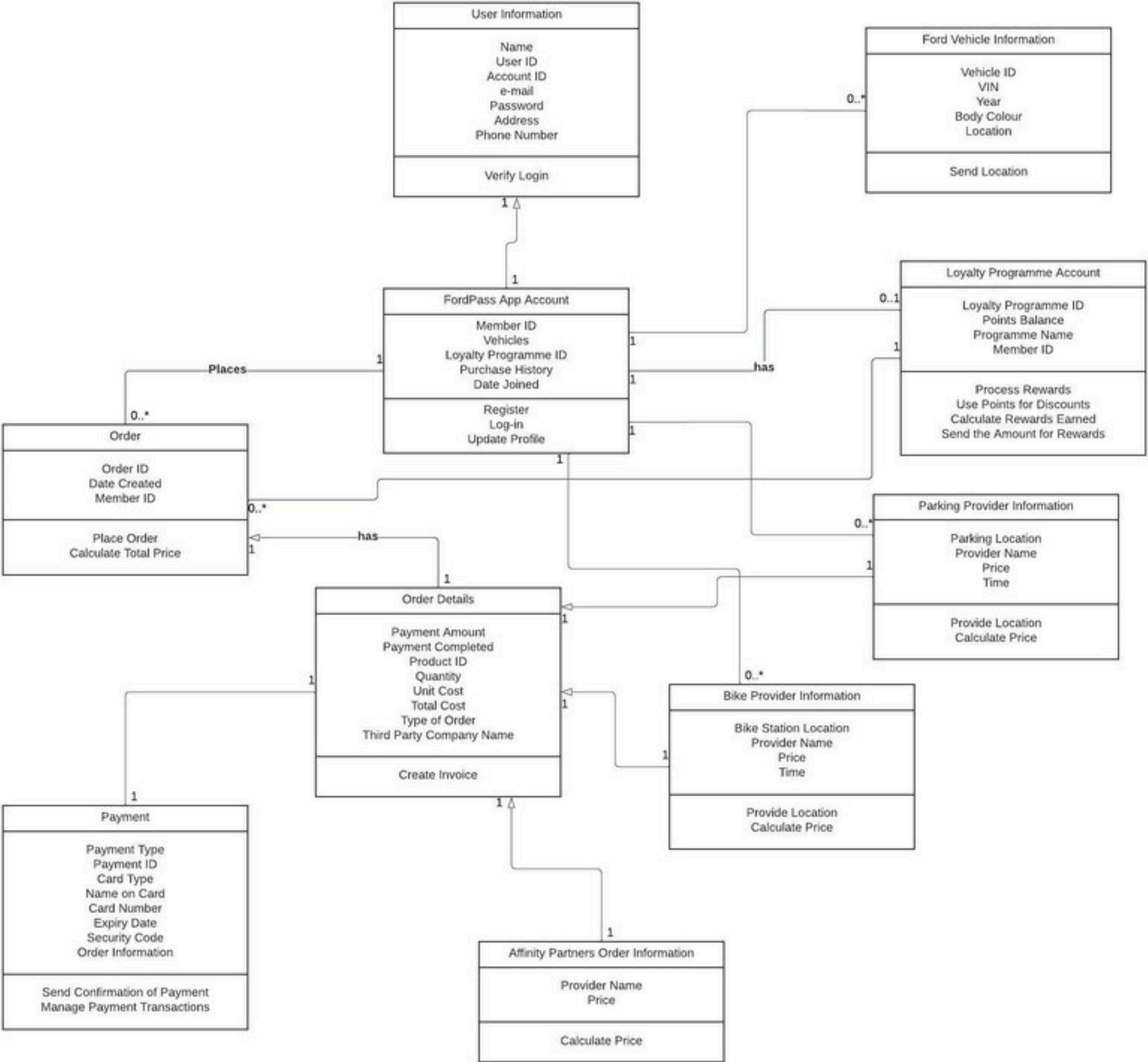
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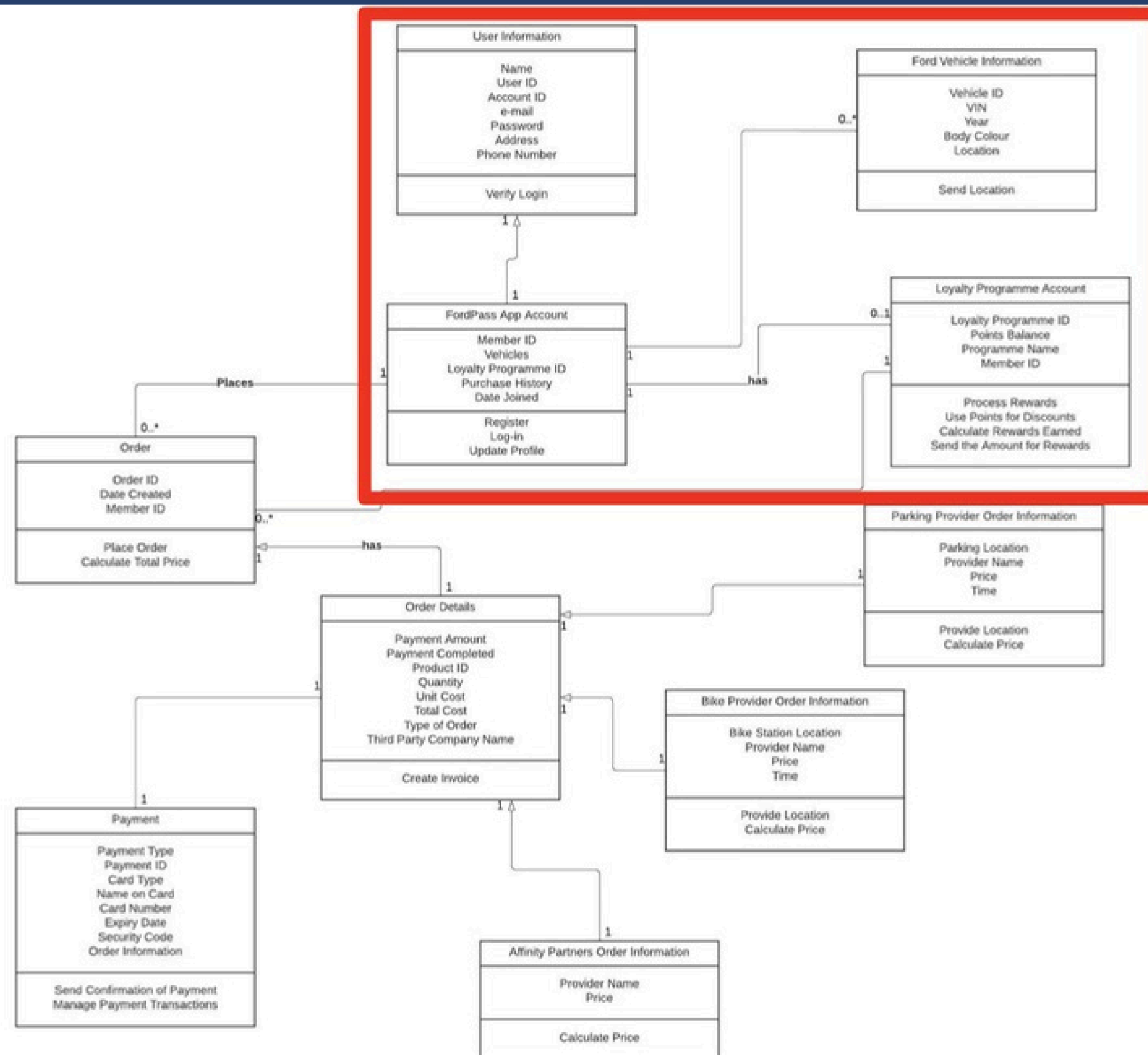
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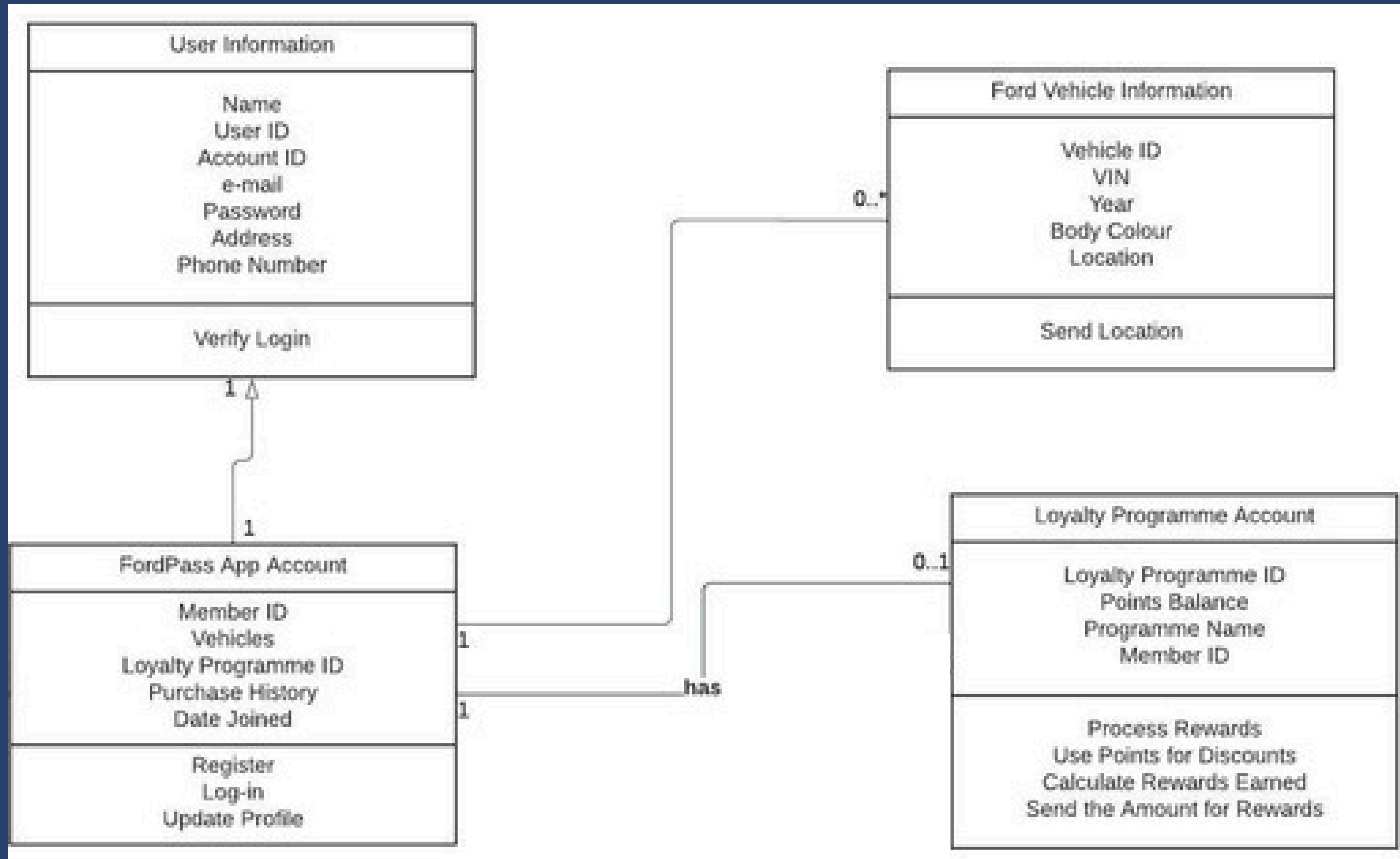
BUSINESS CLASS DIAGRAM

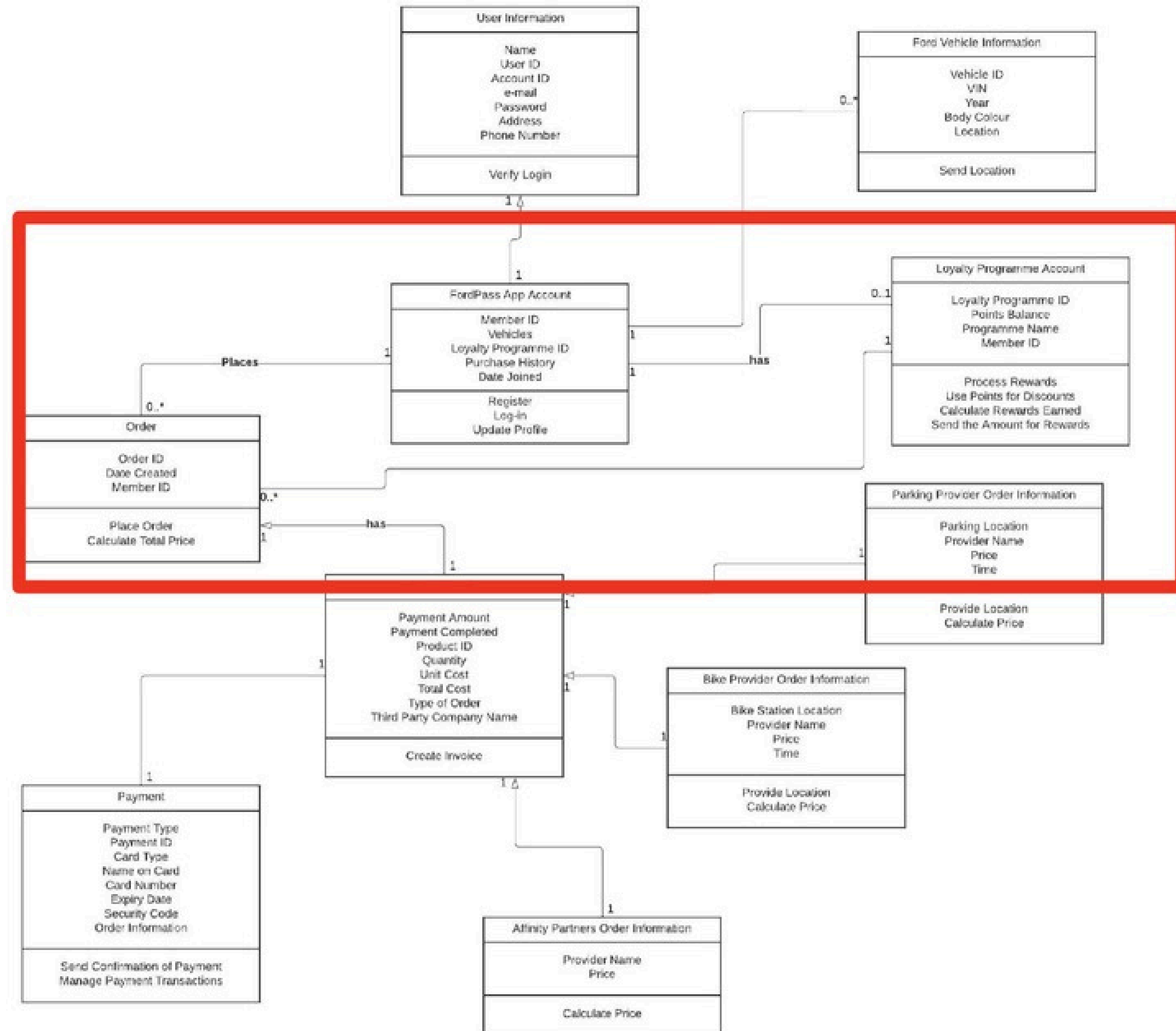


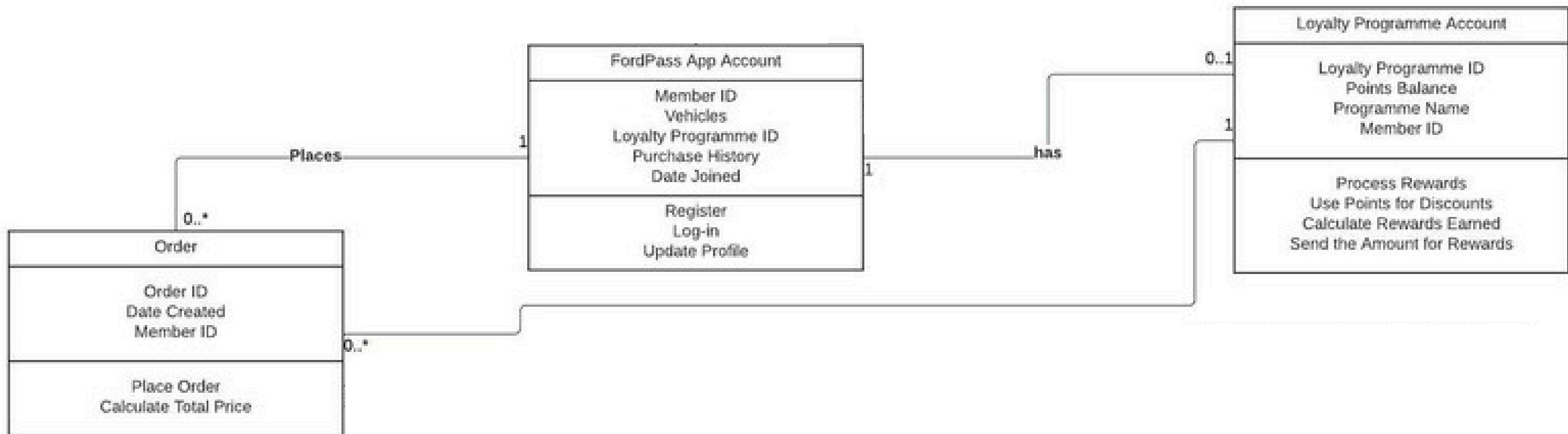
FordPass App Account
Member ID Vehicles Loyalty Programme ID Purchase History Date Joined
Register Log-in Update Profile

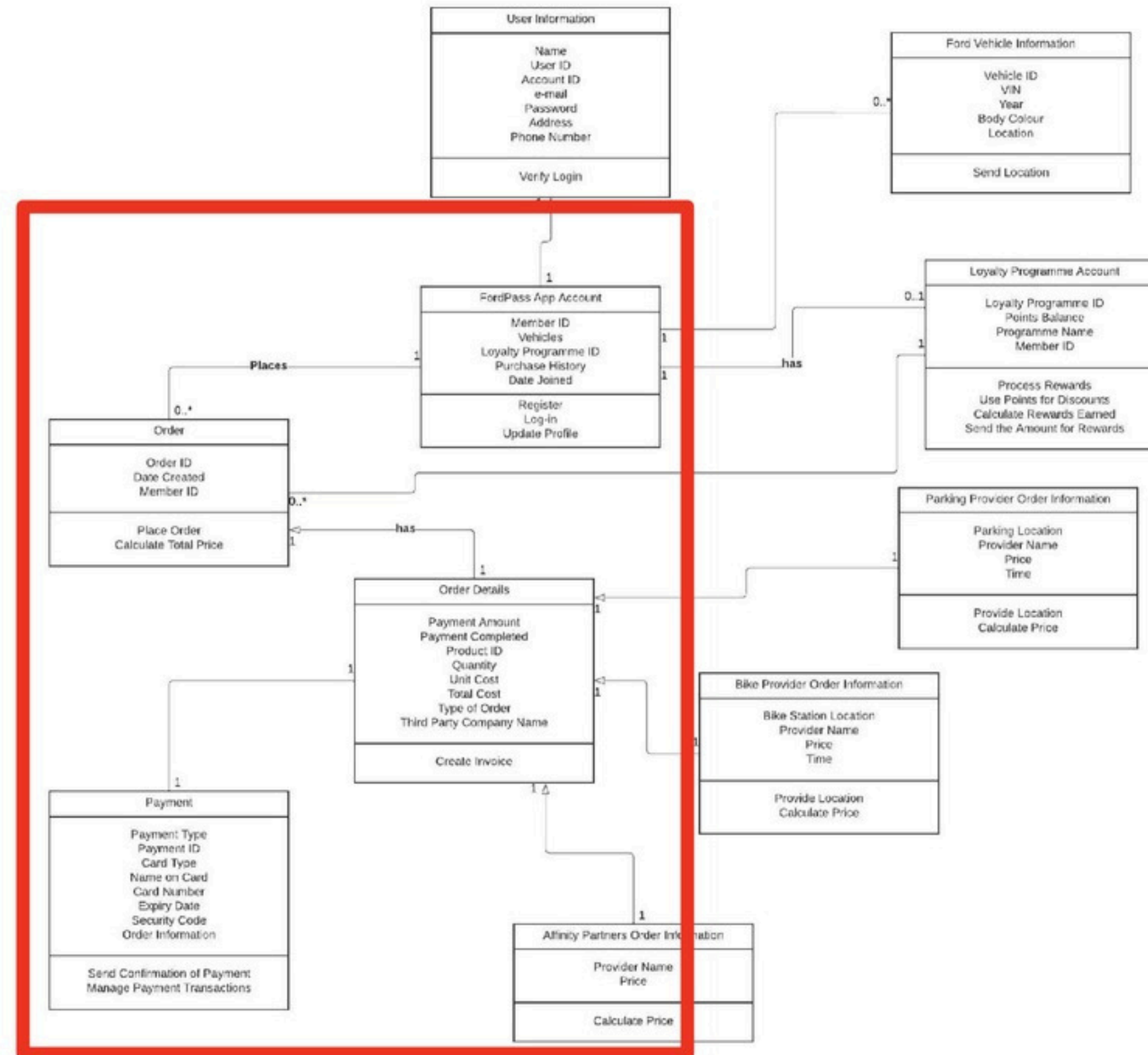


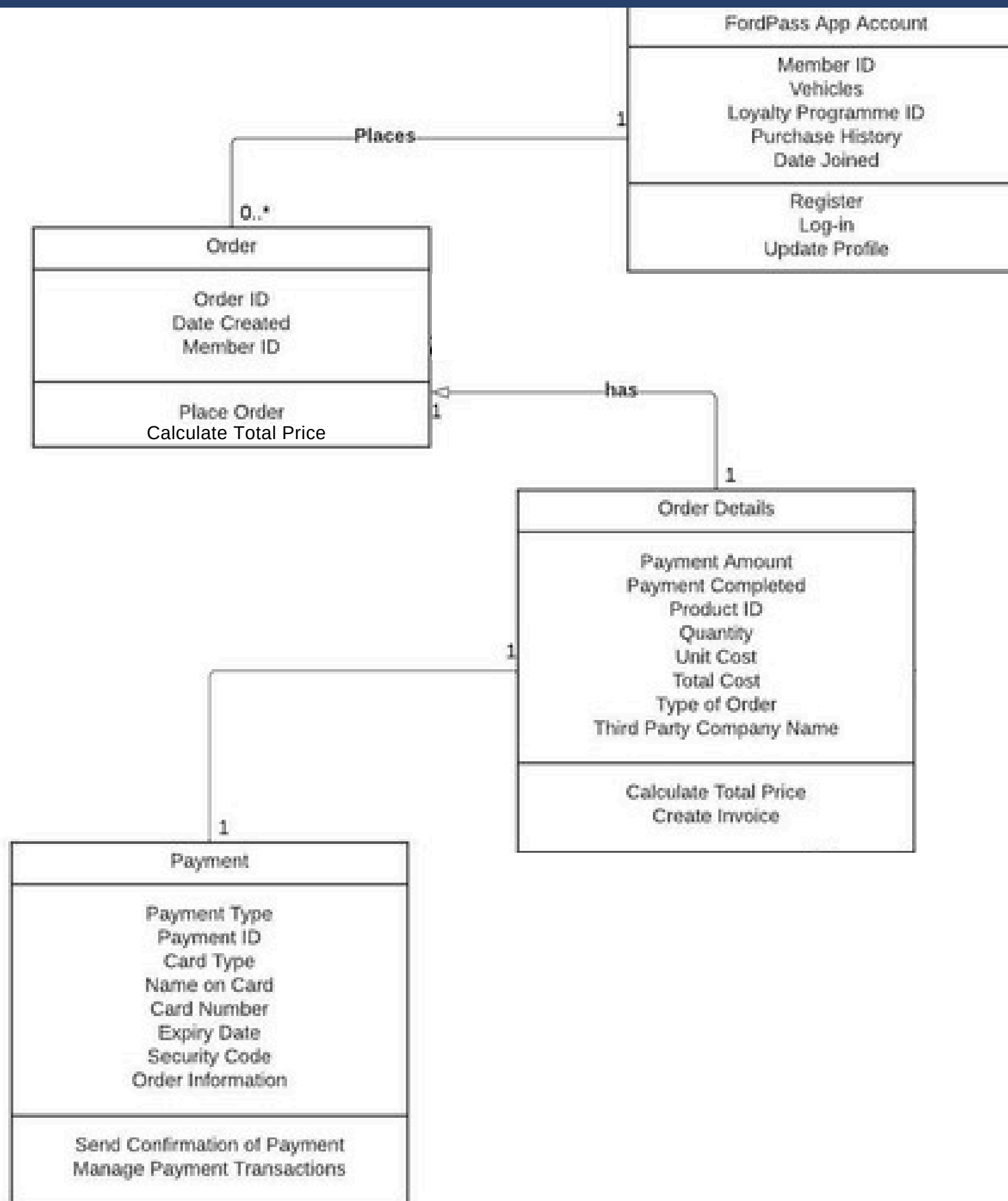


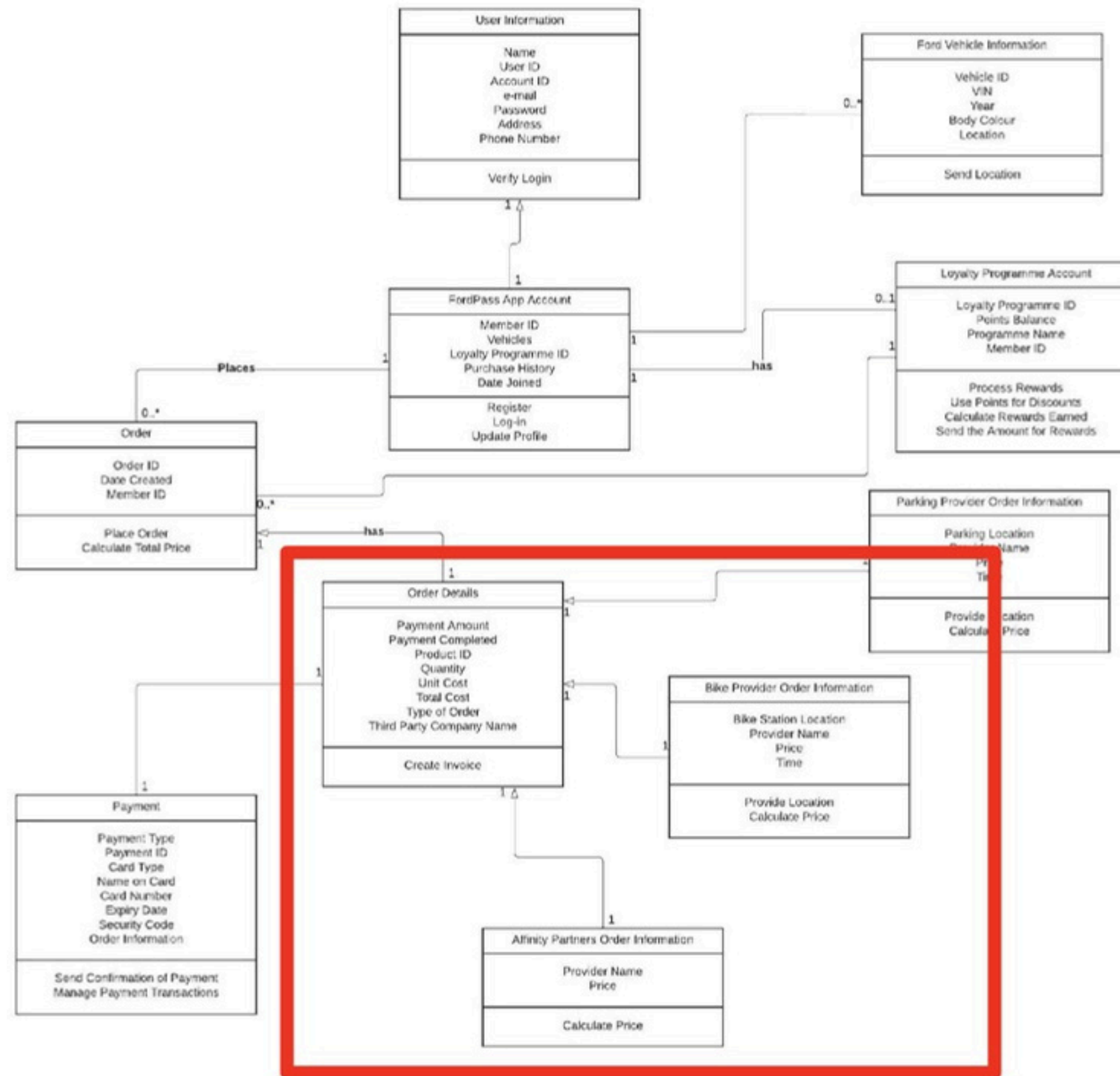


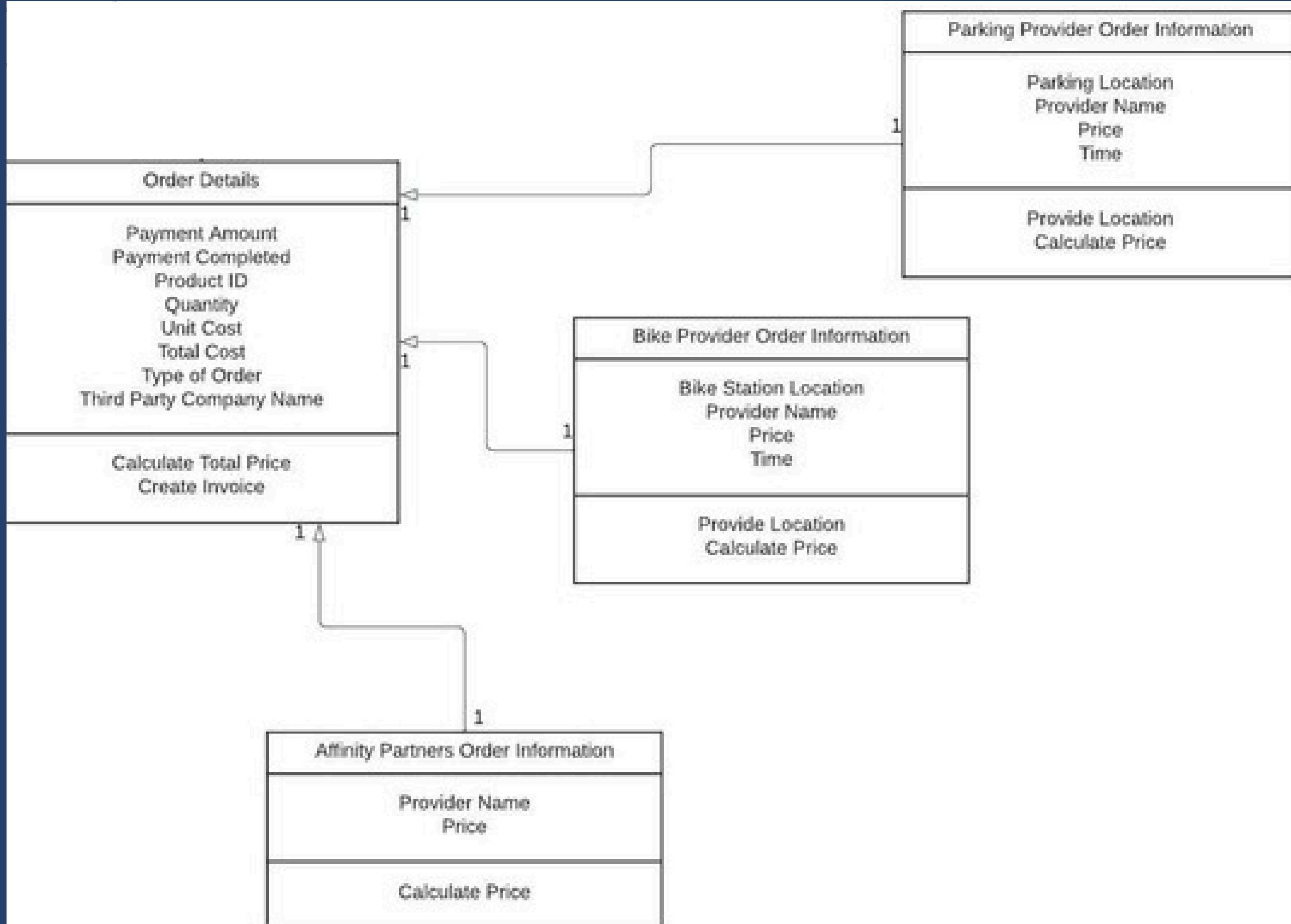












Brief Understanding

DATA FLOW DIAGRAM

*Visual representation of the
flow of data
through a process or
system.*

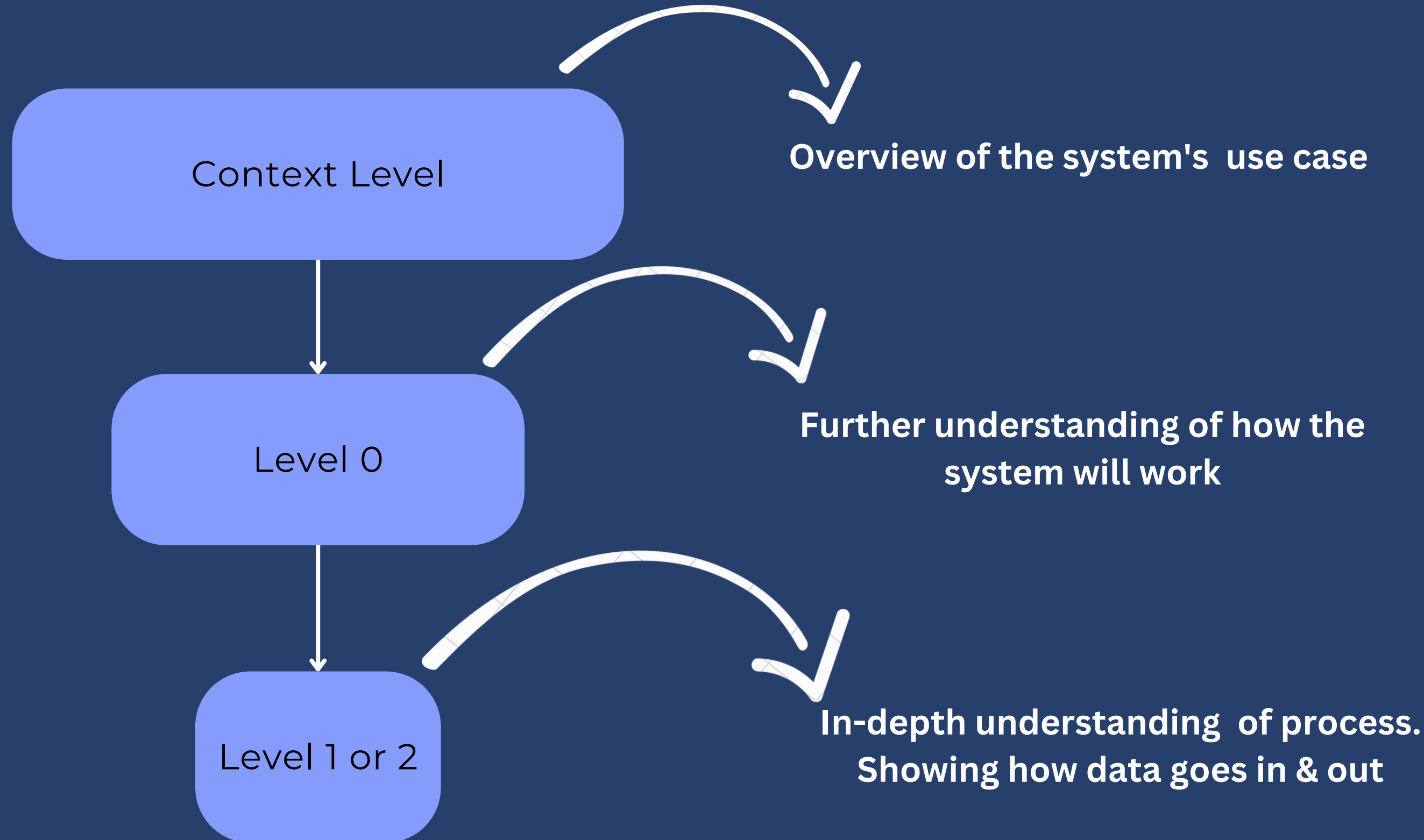


Diagram Understanding

DATA FLOW DIAGRAM

*Visual representation of the
flow of data
through a process or
system.*



External Entities

Process

Data Storage

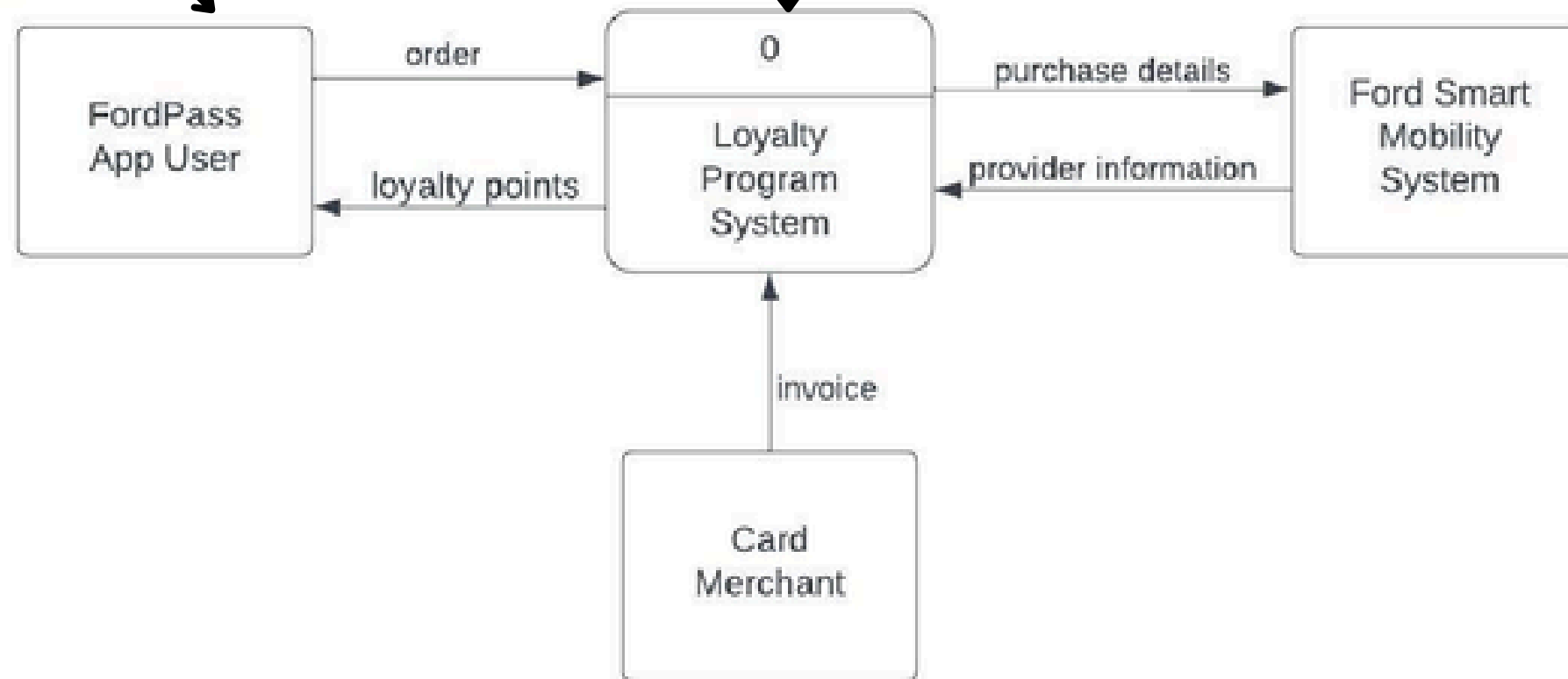
Data Flow

Entities

UseCase

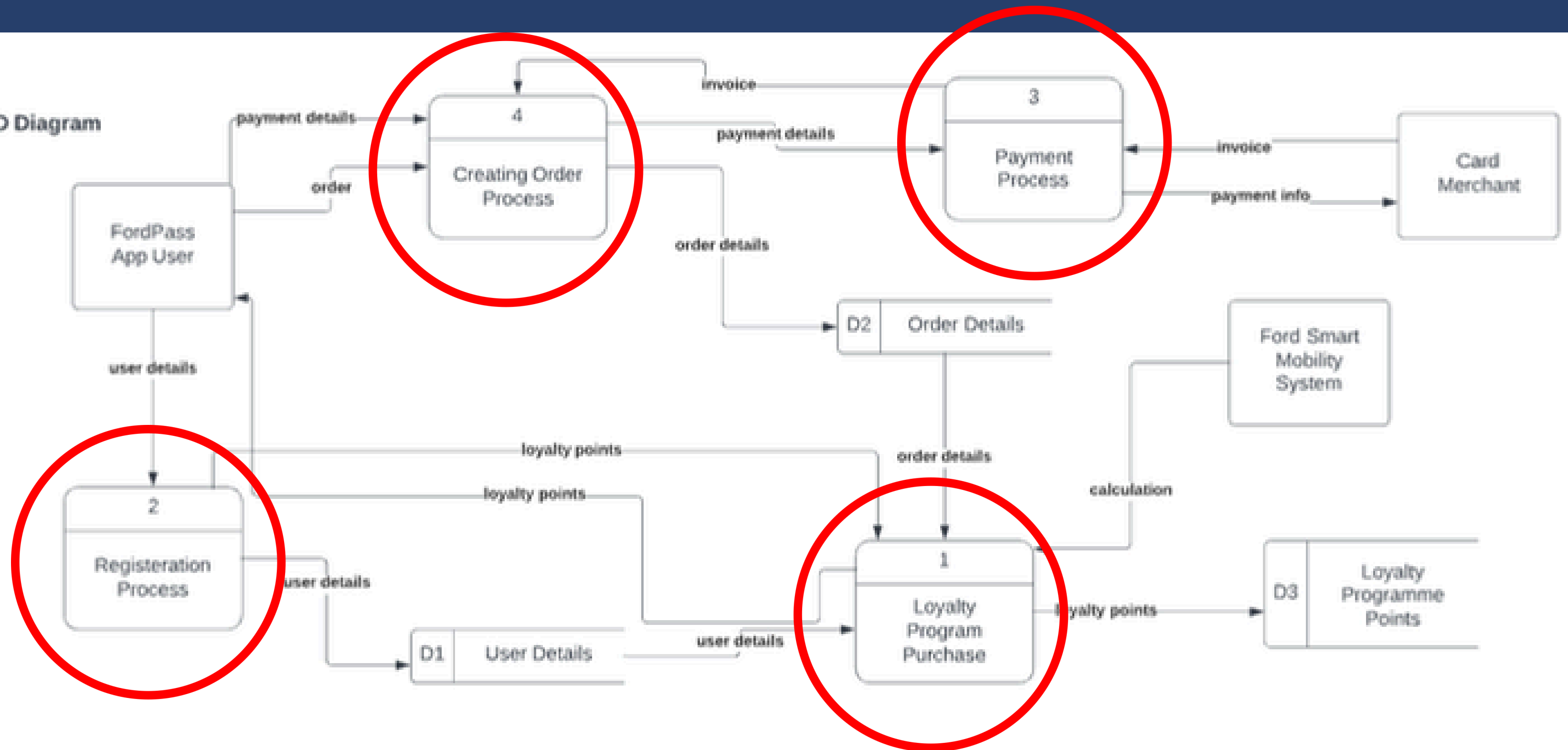
LOYALTY PROGRAM

Context Diagram



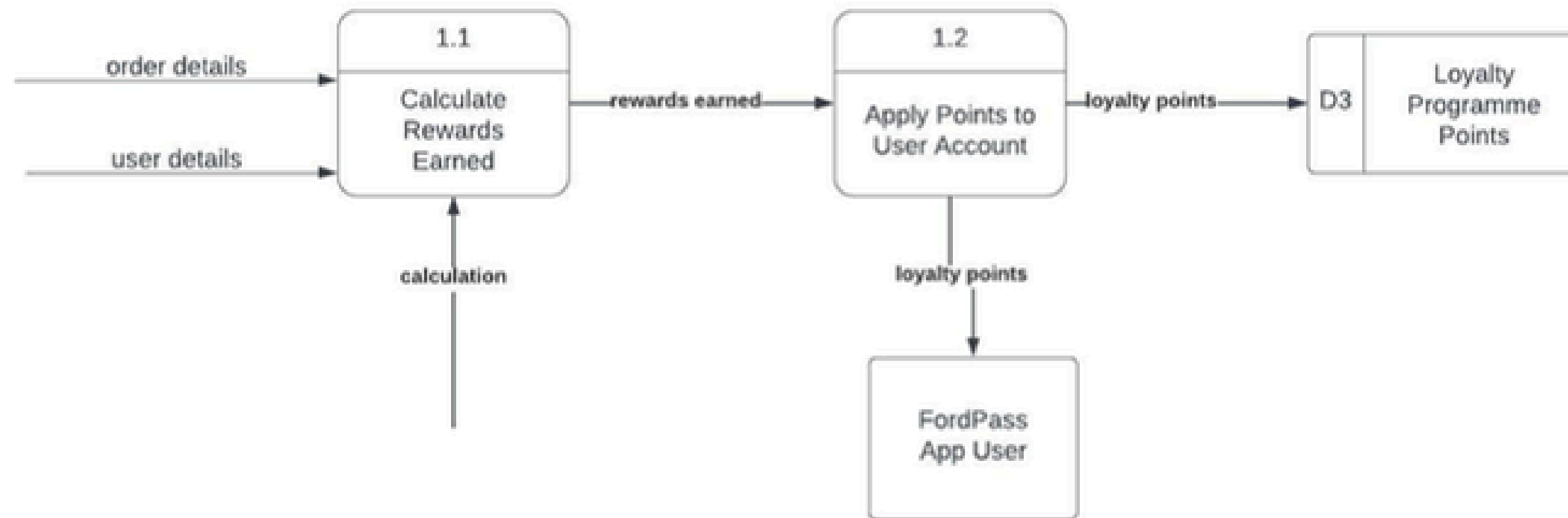
LOYALTY PROGRAM

Level 0 DFD Diagram



LOYALTY PROGRAM

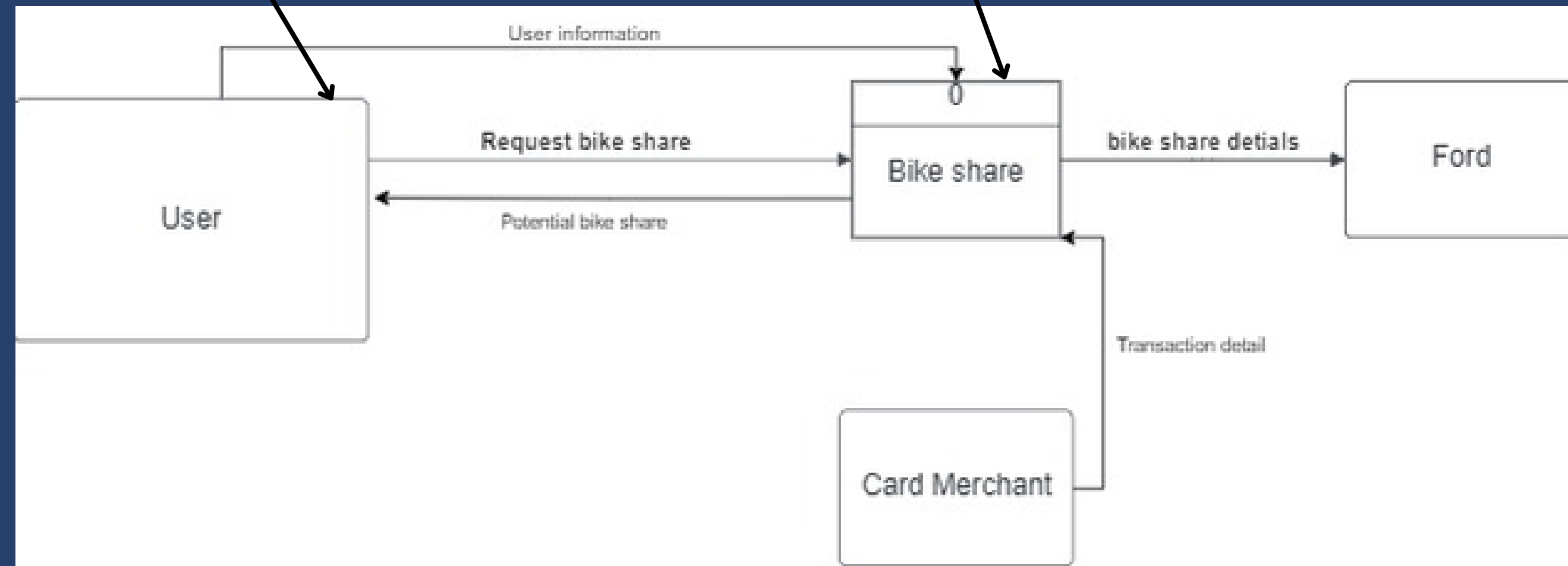
Level 1 DFD Diagram
Process 1: Loyalty
Program Purchase



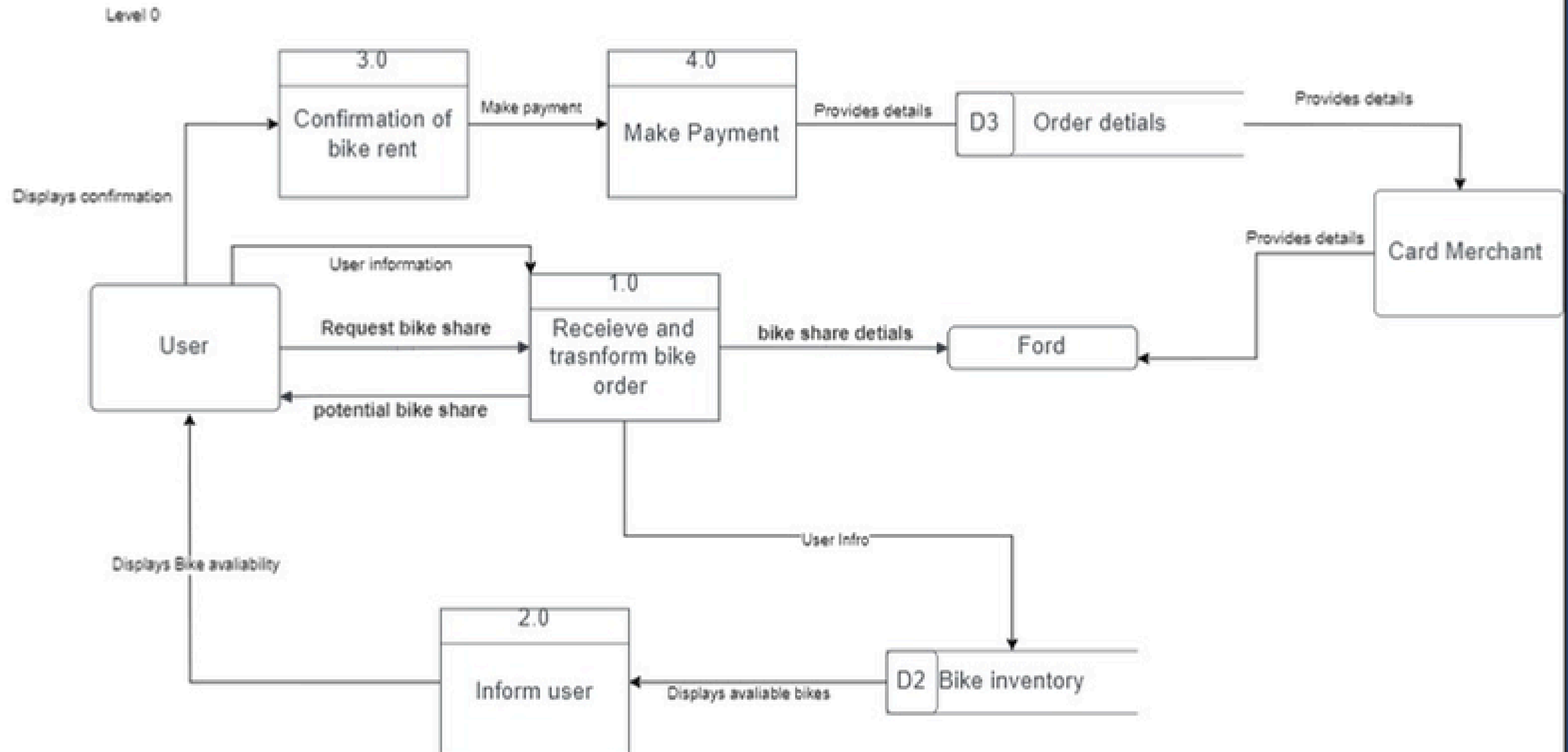
Entities

Use Case

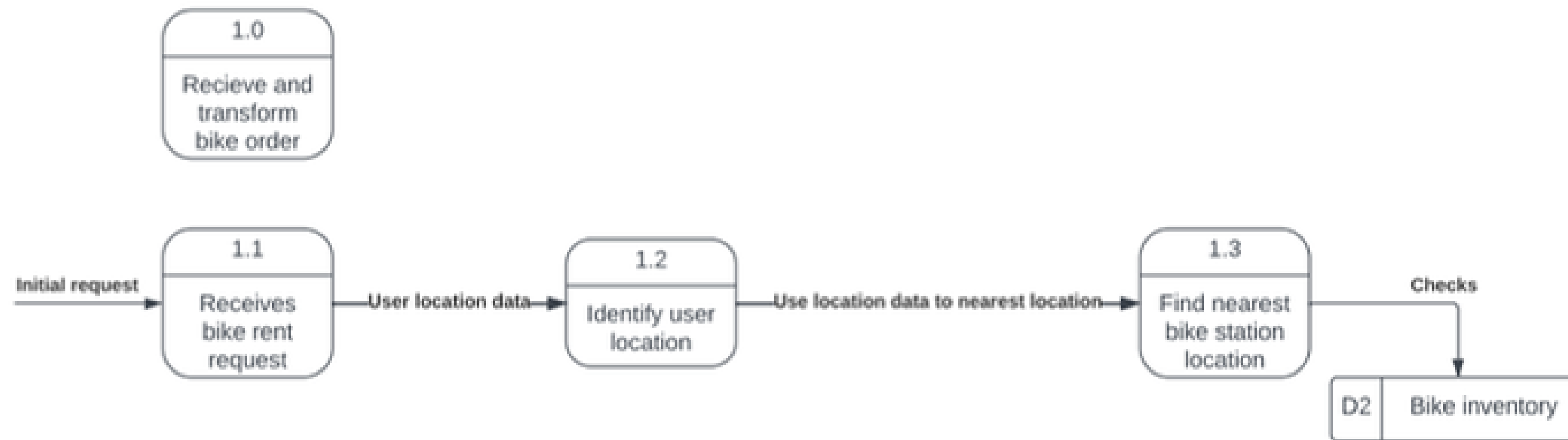
BIKE SHARE



BIKE SHARE



BIKE SHARE





QUESTIONS?

BACKUP SLIDES

Use Case Descriptions

Sequence diagram (Bike share).

Sequence diagram (Loyalty program).

Use case diagram (Loyalty program).

Use case diagram (Bike share).

Business class diagram

Data Flow Diagram Loyalty Program

Data Flow Diagram Bike Sharing

Loyalty Program

Bike Sharing

Definitions

Use Case Diagram

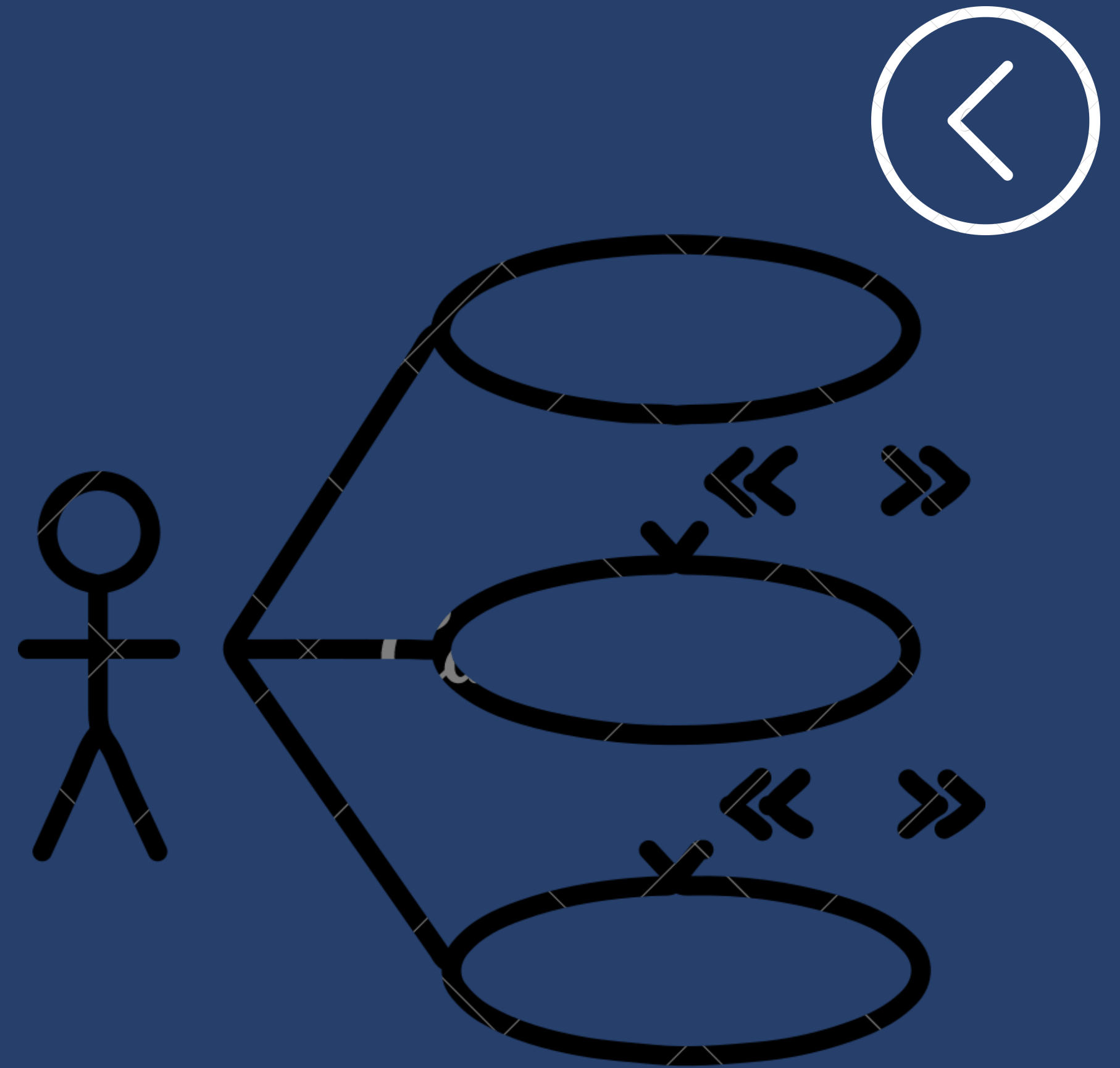
Business Class

Diagram

Sequence Diagram

USE CASE DIAGRAM

A use case diagram is a visual representation of how a system interacts with its users to accomplish specific tasks or goals.



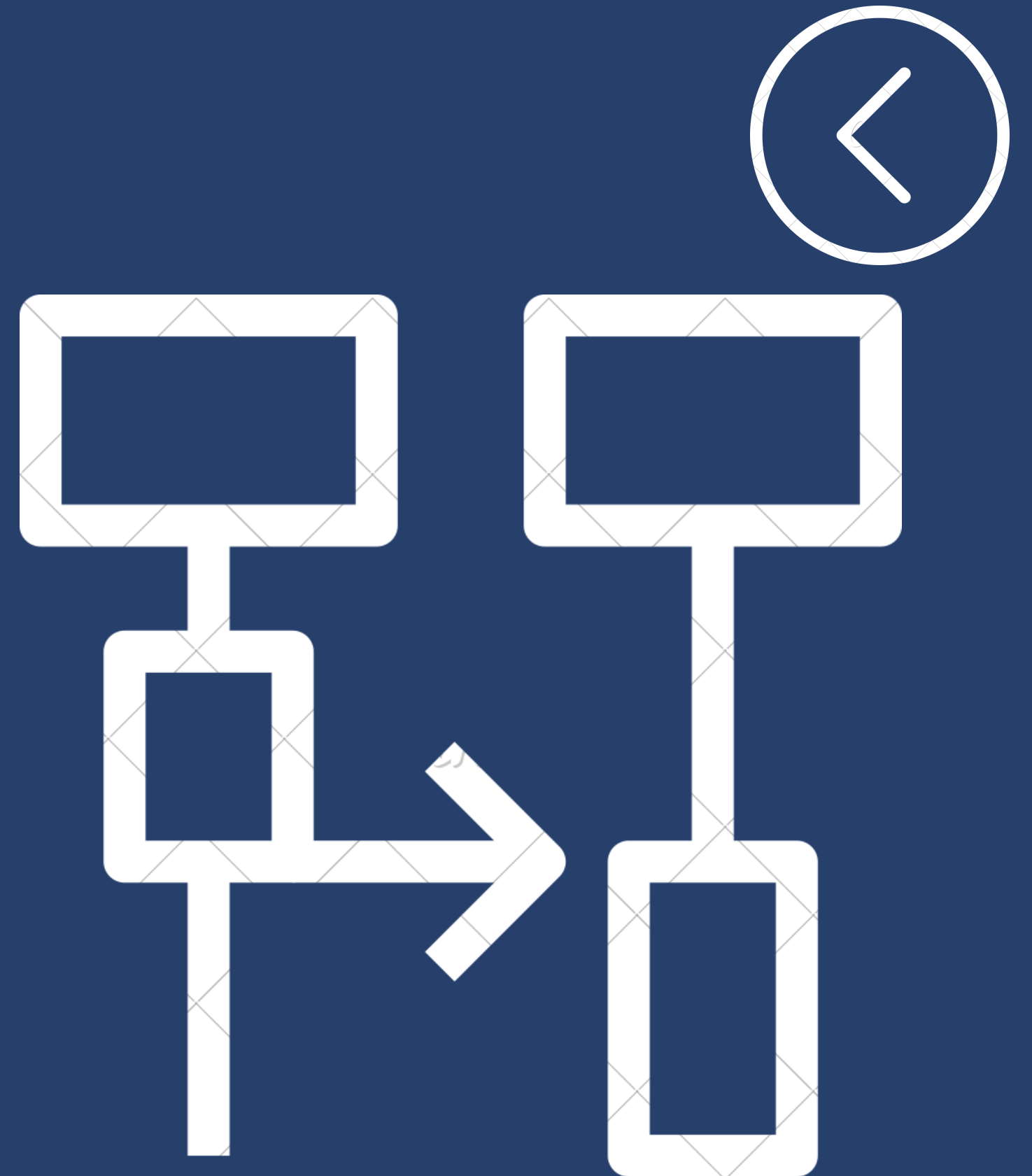
BUSINESS CLASS DIAGRAM

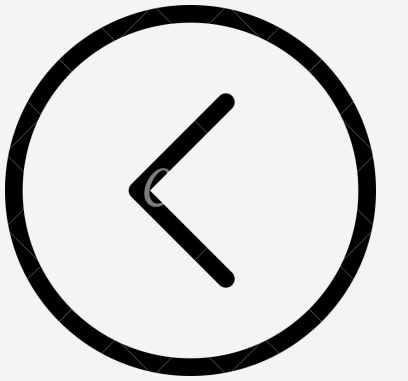
A business class diagram is a type of diagram that shows the high-level relationships between business concepts or entities in a system.



SEQUENCE DIAGRAM

A sequence diagram is a type of diagram in software engineering that shows the interactions between objects or components in a system over time.





Loyalty Program

Use Case:

Reserve Parking

Pay in Advance for Parking

Get Registered

Add Vehicle

Receive Members only Bonuses

Interact with Ford Products and Services

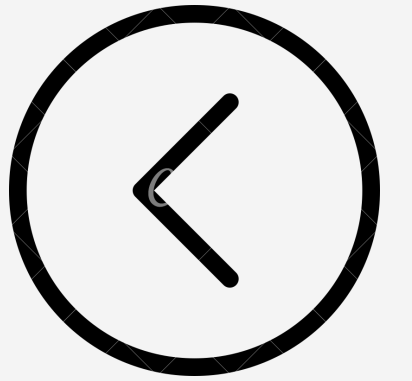
Collaborate with Affinity Partners

Give Rewards to Members

Send Parking Information to Member

Process Payment

Bike share



Use case

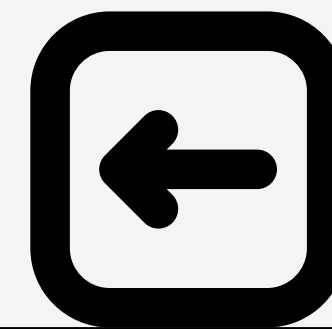
Login

Find bike from nearest pickup station

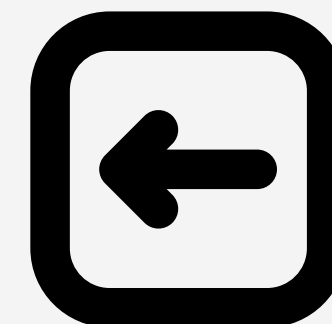
Reserve bike

Display nearest bike station

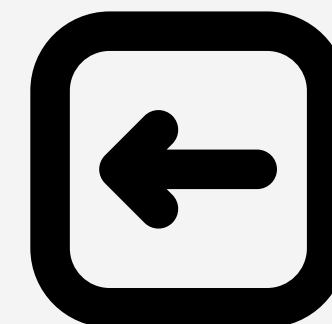
Get bike pickup details



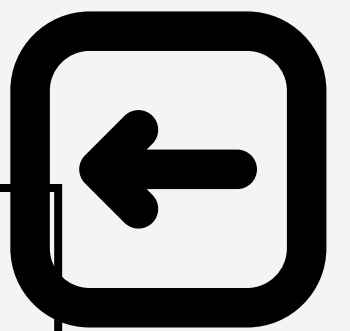
Use Case:	Reserve Parking
Goal:	Parking their car
Sequence:	A. Members open the app. B. Member signs in. C. Members search for a parking space. D. FordPass App delivers the closest parking spaces and prices. E. Members view the closest parking spaces and their prices. F. Members reserve a parking space. G. FordPass App confirms the reservation and informs the car park. H. Car park ensures a spot has been reserved. I. Confirmation details have been saved to the FordPass App. J. Reward points have been added to the FordPass App. K. Member parks on site. L. Member leaves.
Alternatives:	At B, if the user has no account; they can register. At F, the member can choose to pay for parking in advance. At K, if the member has not paid by this point, use ‘Pay in advance for parking’ Use Case.



Use Case:	Pay in advance for parking
Goal:	Pay for Parking in Advance
Sequence:	A.) Use “Reserve Parking” B.) Use CaseMembers pick location and parking spots. C.) Member clicks pay now.Payment page appears. D.) Customer inputs card details or uses a saved card info. E.) Payment provider confirms payment and informs Ford PassApp. F.) FordPass App confirms payment. G.) Use ‘Receive Rewards’ H.) Use Case.Payment is lodged under ‘Bookings’.
Alternatives:	At B, if the user has no account; they can register. At G, members can reserve now and pay later.

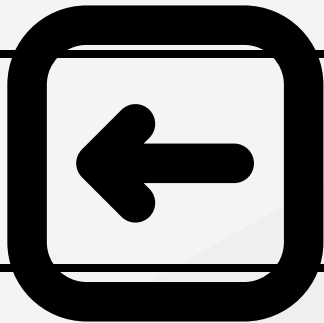


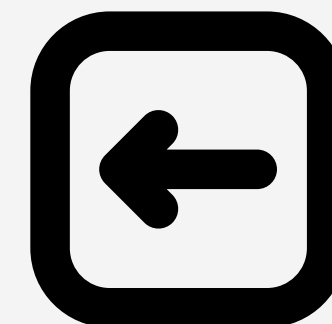
Use Case:	Get Registered
Goal:	To become a FordPass App member
Sequence:	A. User opens app store on mobile device B. Users download FordPass App. C. User clicks register. D. User inputs personal details to create an account - name, email address, password. E. Member authorises permissions on device -location, access to eWallet. F. Member inputs details to best utilise the app - car registration, favourite car park, most visited locations. G. Members are fully registered.
Alternatives:	



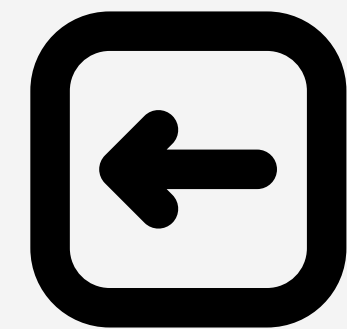
Use Case:	Add vehicle
Goal:	Add a vehicle to member profile.
Sequence:	A) Member registers for a new account - Use 'Get registered' B) Member clicks on profile C) Member clicks on 'add new vehicle' D) Member enters vehicle details - vehicle registration E) Ford pulls up vehicle information from database F) Ford successfully adds vehicle to member profile.
Alternatives:	At F, if unsuccessful - member would use Ford helpline and they will fix the error.

Use Case:	Purchase Parts or Services
Goal:	Make purchasing products and services easy.
Sequence:	A. Customer opens the app and signs in. B. Customer opens the Purchase parts and services tab C. Customer browses all options available D. Customer searches for service required E. Customer chooses service F. Customer chooses nearest dealership G. Dealership provides date availability H. Customer selects date available and adds to basket I. Customer checks out Payment page appears. J. Customer inputs card details or uses a saved card info. K. Payment provider confirms payment and informs FordPass App. L. FordPass App confirms payment. M. Dealership receives confirmation N. Customer receives confirmation O. Dealer prepares for service P. Customer takes vehicle in for service R. Dealership provide serviceCustomer leaves dealership.
Alternatives:	At D, the customer browses for a part. At E, the customer choses a part and adds it to the basket and goes to step I. At J, if service is free, the customer doesn't need to pay and goes to step O .

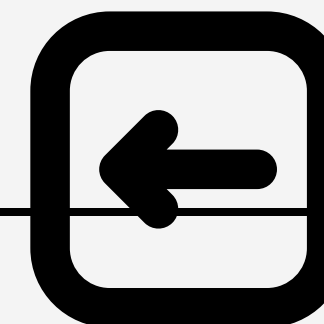




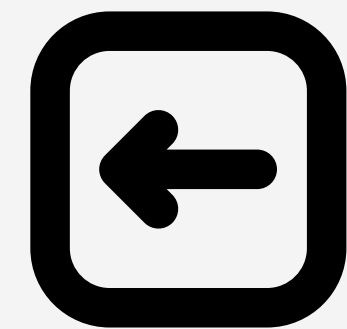
Use Case:	Interact with Ford products and services
Goal:	Enticing customers to interact with Ford products and services
Sequence:	A.) Member opens app B.) Member signs in C.) Member scrolls through app - views, products and services D.) Member favourites products and services
Alternatives:	



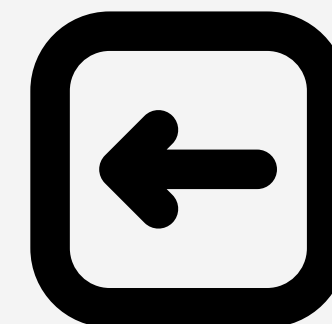
Use Case:	Collaborate with Affinity Partners
Goal:	Purchasing from affinity brands
Sequence:	A.) Ford agrees a deal with affinity partners. B.) Members use applications and receive rewards. C.) Ford app advertises affinity partners. D.) Members go to an affinity partner and choose an item(s) to purchase. E.) Use 'Recognize Members' F.) Use CaseCustomers purchase item(s) at a respective discounted rate. G.) Ford Pass deducts rewards spent. H.)Ford app promotes more possible offers. I.) Use CaseUse 'Receive members only bonus'.
Alternatives:	



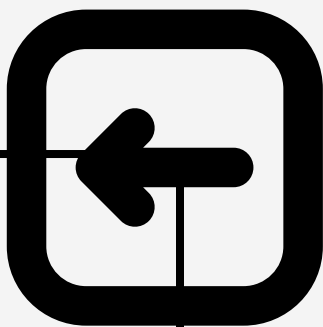
Use Case:	Give rewards
Goal:	Give rewards to member profile.
Sequence:	A) Member signs in. B) Member reserves parking - use 'Reserve parking' Use Case. C) Payment confirmation has been received. D) Points are calculated against cost spent by member. E) Points are added to member profile F) Members are notified of point changes. G) Points are changed into discounts upon accessing by affiliate brand. H) Use 'Collaborate with affinity partners'
Alternatives:	At B, member may pay in advance - use 'Pay in advance for parking' Use Case.



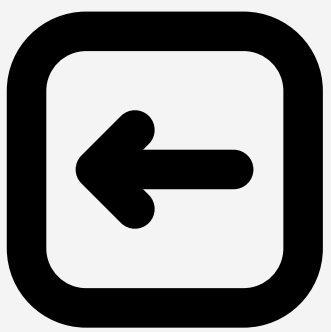
Use Case:	Sending parking information to member
Goal:	Sending parking information to member
Sequence:	A. Use 'Reserve parking' B. Use CaseUse 'Process payment' C. Card merchant sends confirmation of payment D. Send confirmation of payment to customer E. Send booking details to car park. F. Send booking details to customer - date, time, balance, G.location, booking reference
Alternatives:	



Use Case:	Process Payment
Goal:	Taking a payment from customer
Sequence:	A. Member signs in B. Member selects parking C. Payment window shows D. Customer inputs card details E. Wait for confirmation from bank that payment has been accepted F. Sends confirmation of payment to Ford and member.
Alternatives:	At A, if user is not a member yet use 'Get registered' Use Case. At B, 'Purchase parts or services' Use Case if customer does not wish to book for parking.

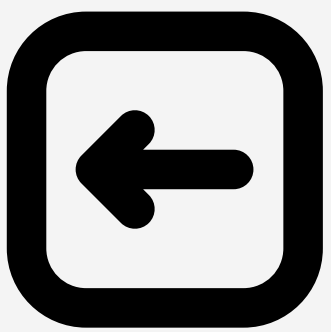


Use Case:	Log-in to Application
Goal:	Renting a bike
Sequence:	A. Members open the app B. Member signs in C. Members search for a bike. D. FordPass App delivers the closest bike stations and prices. E. Members view the closest bike station and their prices. F. Members reserve a bike by giving card details to the ford pass app. G. FordPass App confirms the reservation. H. fordPass app sends the details to the payment provider I. Confirmation details have been saved to the FordPass App. J. Members get the bike. K. Members return the bike after trip L. Members get charged on their card for the hour.
Alternatives:	At B, if the user has no account; they can register. At F, the member can choose not to rent the bike.

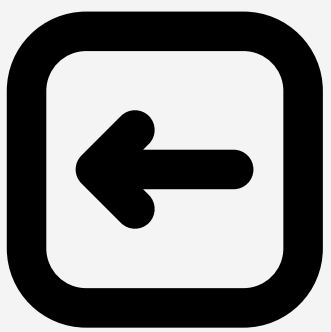


Use Case:	Find bike from nearest pickup station
Goal:	Find a bike at the nearest location
Sequence:	A) Member searches for bike B) Ford pass app finds the nearest bike location C) Ford pass notify users of the nearest bike location D) Fordpass request user to confirm rent
Alternatives:	At B, if FordPass can't find a bike. Showing "No bike available" at C. At D, if the user does not confirm the rental bike. Showing another nearest bike or asking whether to cancel or not.

Use Case:	Park the bike at the destination bike station
Goal:	Getting users to park at the bike station.
Sequence:	A) Members go on the FordPass app to see where to park bikes. B) FordPass app finds the nearest bike station C) FordPass app notifies users of the nearest bike station D) Fordpass App request whether user wants to return their bikes to that station
Alternatives:	At D, if the user does not confirm they want to return their bikes. Then cancel the whole process until users search for the nearest bike station.



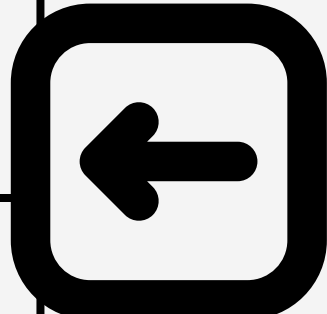
Use Case:	Reserve bike
Goal:	Users reserve a bike.
Sequence:	A) Members go on the fordpass app to reserve a bike. B) Fordpass app finds the nearest bike station C) Fordpass app notifies users of the nearest bike station. D) Fordpass app finds a bike. E) Fordpass app shows estimated rent for an hour. F) Members choose to reserve after seeing the estimated hourly rent. G) Members are requested to provide the card details to reserve. H) Members scan card I) Members enter pin J) Card details sent to the payment provider. K) The payment provider confirms the card pin. L) Members complete the reservation.
Alternatives:	At B, fordpass app may not find any nearest station. At D, fordpass may not find any available bike. At E, members may decide not to go ahead with renting. At H, the payment provider may not be able to verify the card.



Use Case:	Display nearest bike station
Goal:	Users need a nearest bike station to find a bike.
Sequence:	a. Members go on the fordpass app to reserve a bike. b. Members put their location in the fordpass app. c. Fordpass app finds the nearest bike station. d. Fordpass app notifies users of the nearest bike station.
Alternatives:	At c, the fordpass app may not find the nearest bike station.

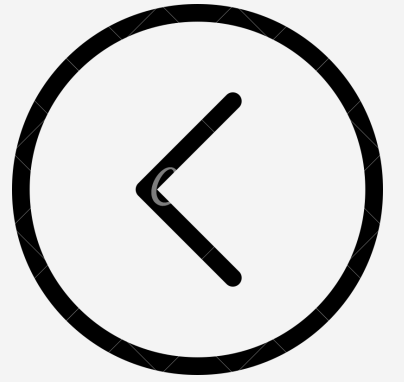
Use Case:	Get customer card details
Goal:	To reserve bike
Sequence:	a) Members choose a bike to reserve. b) Fordpass app shows estimated rent for an hour. c) Members choose to reserve after seeing the estimated hourly rent. d) Members are requested to provide the card details to reserve. e) Card details sent to the payment provider. f) Card details are verified by the payment provider. g) Payment provider confirms card details. h) Members complete the reservation.
Alternatives:	At c, members may not choose to go ahead with the reservation. At d, members may not provide card details. At f, members may fail verification.

Use Case:	Get bike pickup details
Goal:	Getting to the location of the bike station.
Sequence:	A) Members reserve a bike through fordpass. B) Members are given a confirmation of the reservation. C) Members are shown the location and the distance to the bike station.
Alternatives:	At A, Members may not reserve a bike.

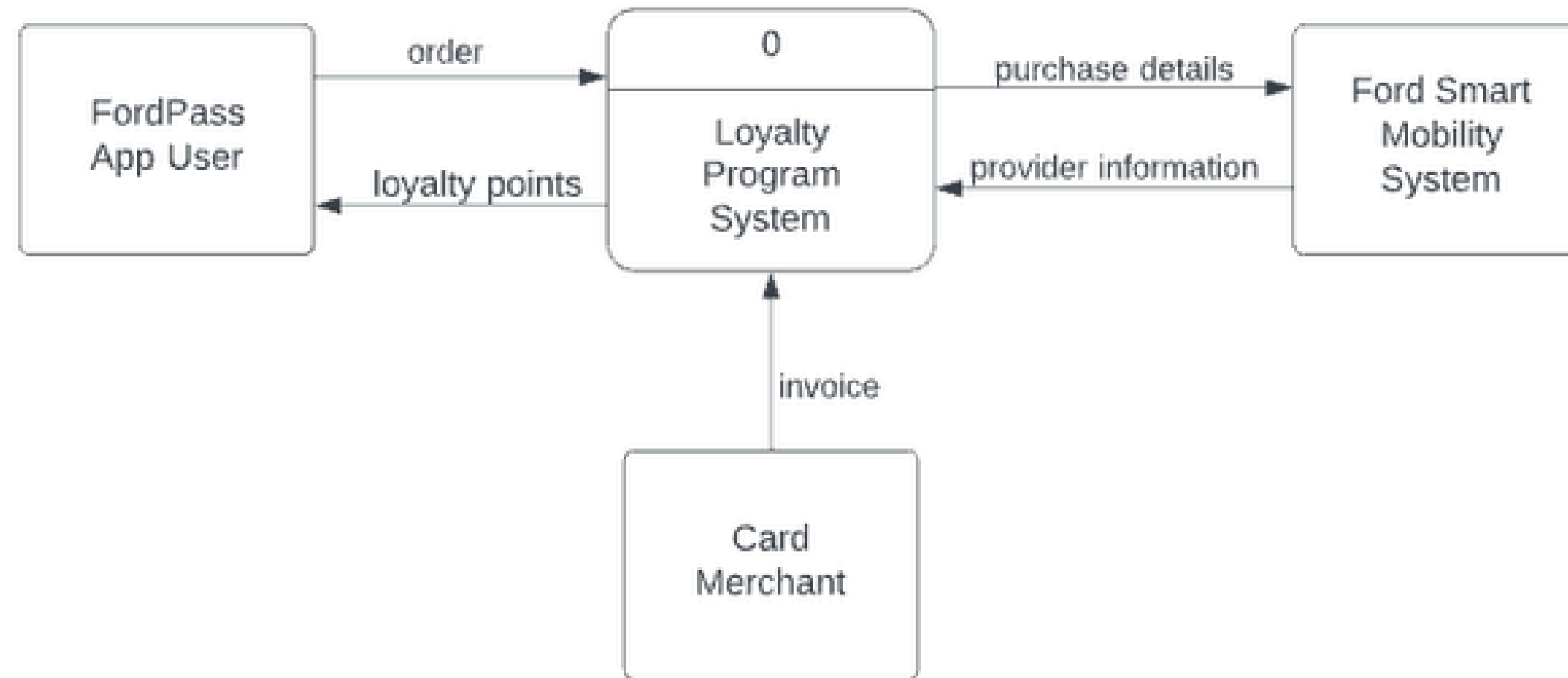


Use Case:	Get bike destination details
Goal:	Getting to the chosen destination of the user.
Sequence:	A) Members choose their destination by using fordpass. B) fordpass shows the nearest bike station. C) Fordpass asks for card details to reserve. D) Members provide card details E) card details are verified. F) Fordpass confirms destination. G) Members reserve a bike through fordpass. H) Members are given a confirmation of the reservation.
Alternatives:	At B, there may not be any nearest station. At E, card details may fail verification.

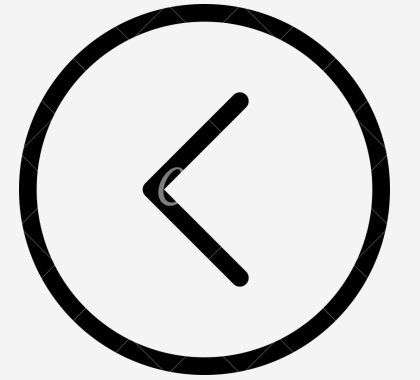
Context Diagram Loyalty Program



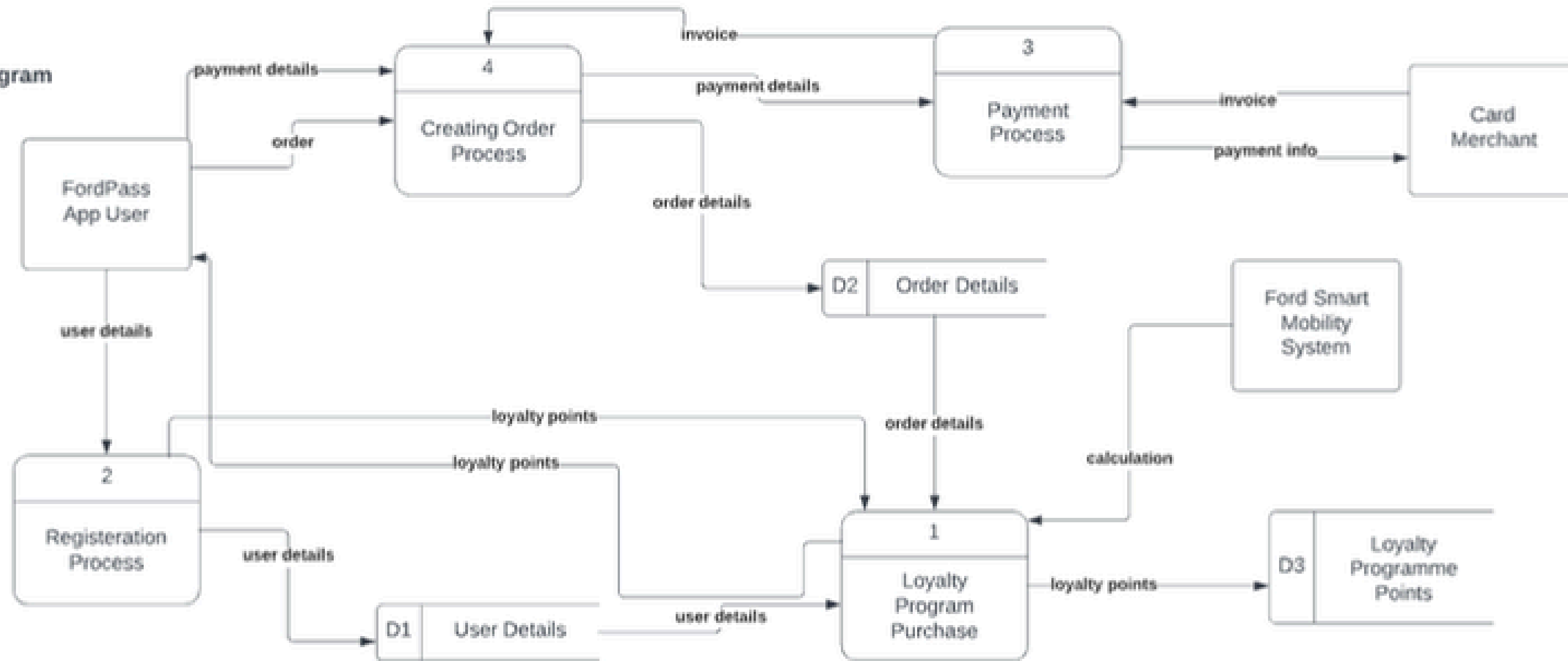
Context Diagram



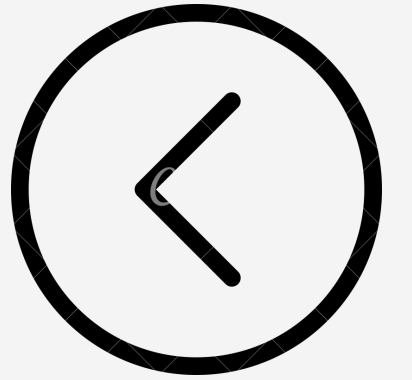
Level 0 DFD Loyalty Program



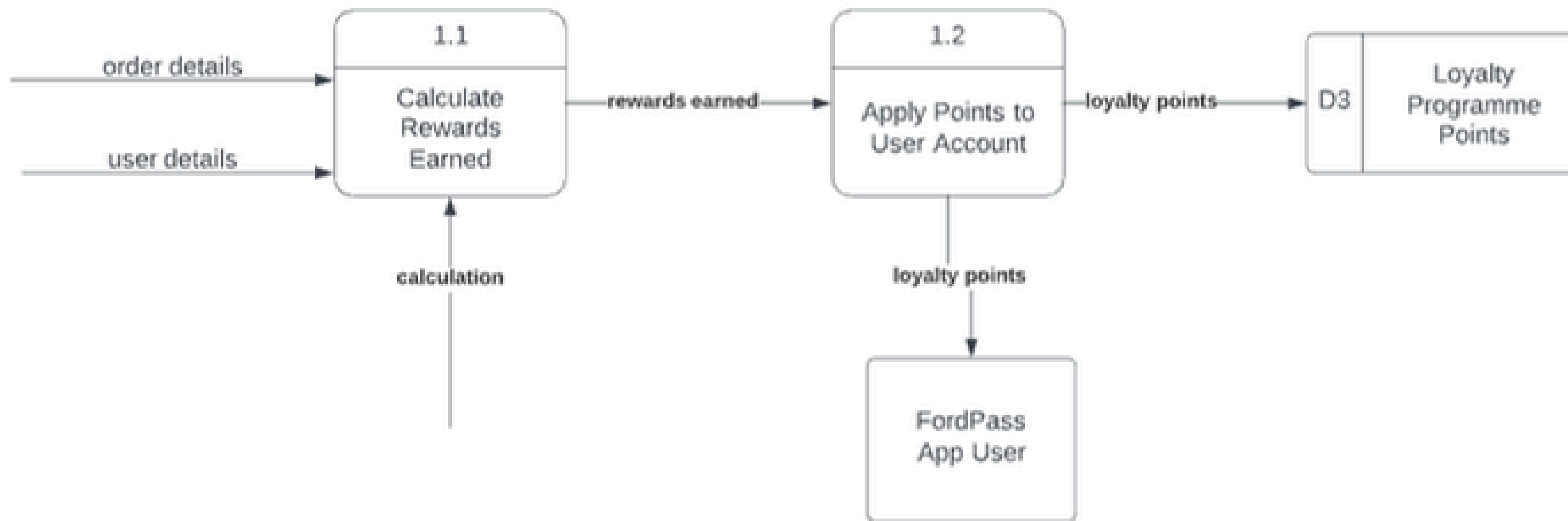
Level 0 DFD Diagram



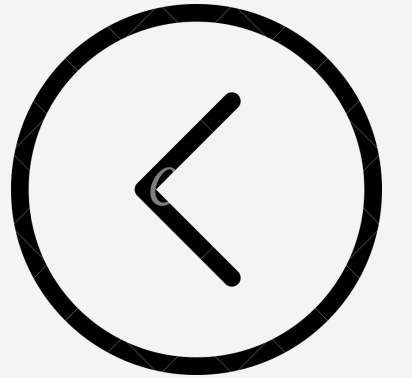
Level 1 DFD Loyalty Program



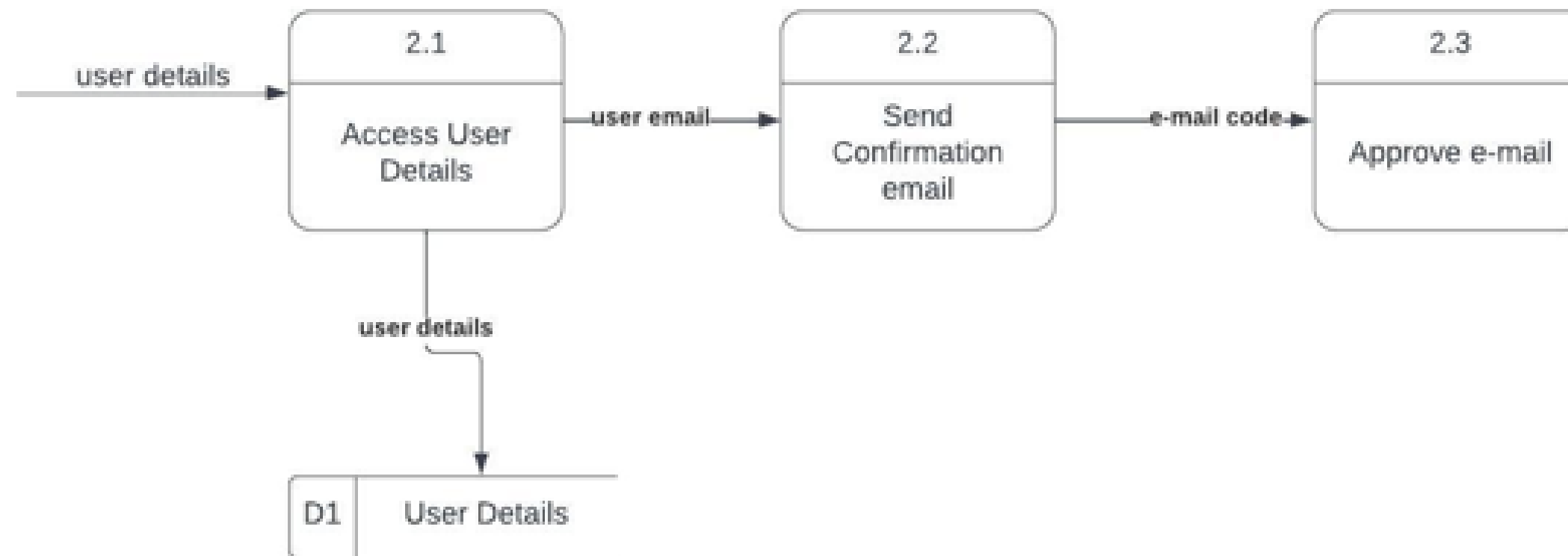
Level 1 DFD Diagram
Process 1: Loyalty
Program Purchase



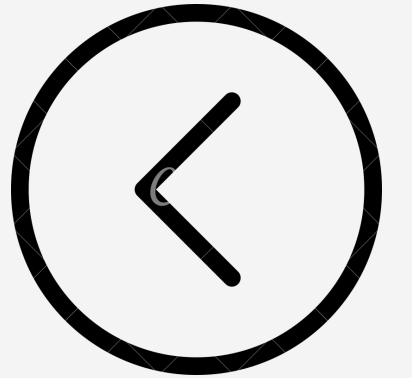
Level 1 DFD Loyalty Program



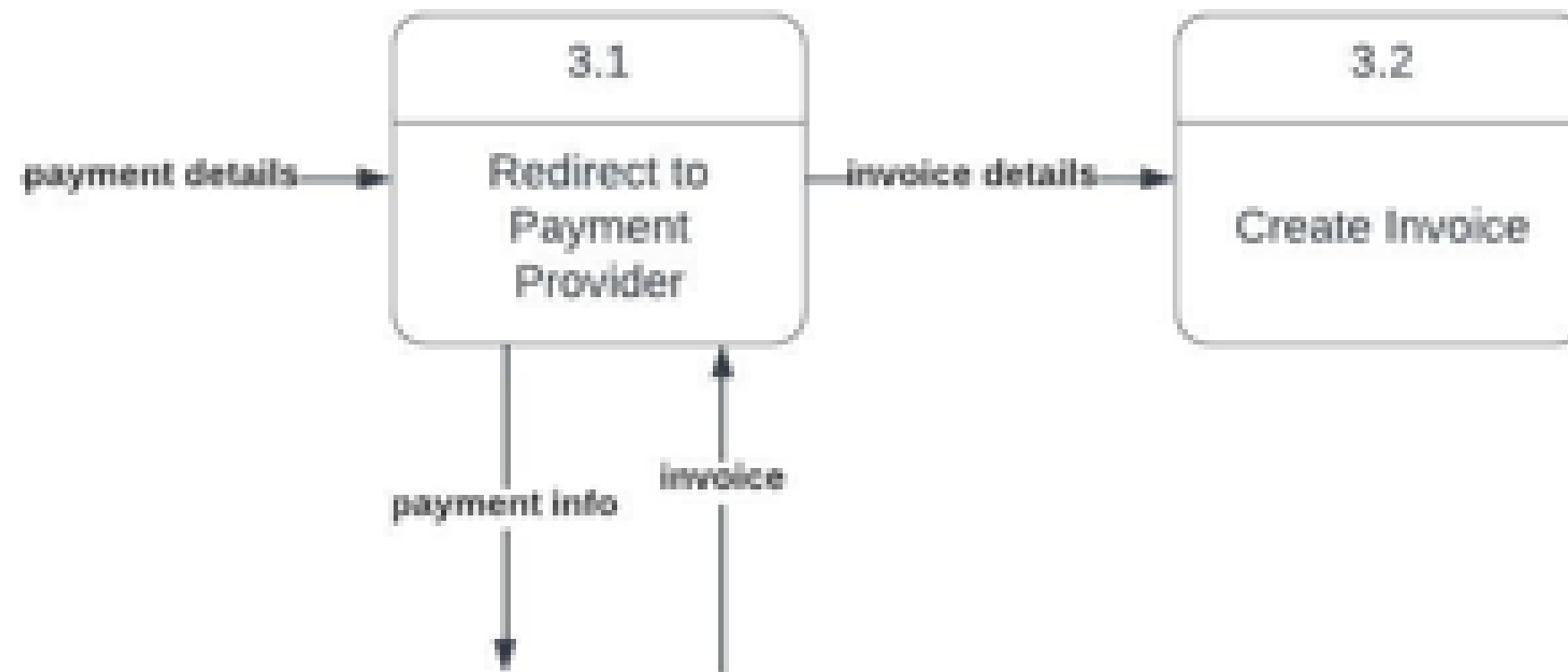
Level 1 DFD Diagram
Process 2: Registration
Process



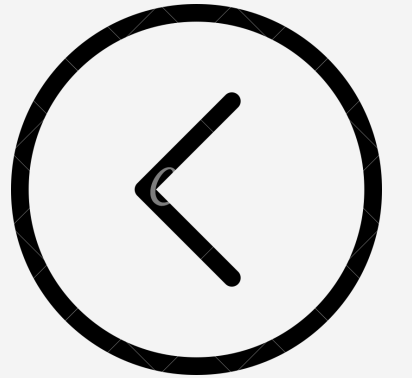
Level 1 DFD Loyalty Program



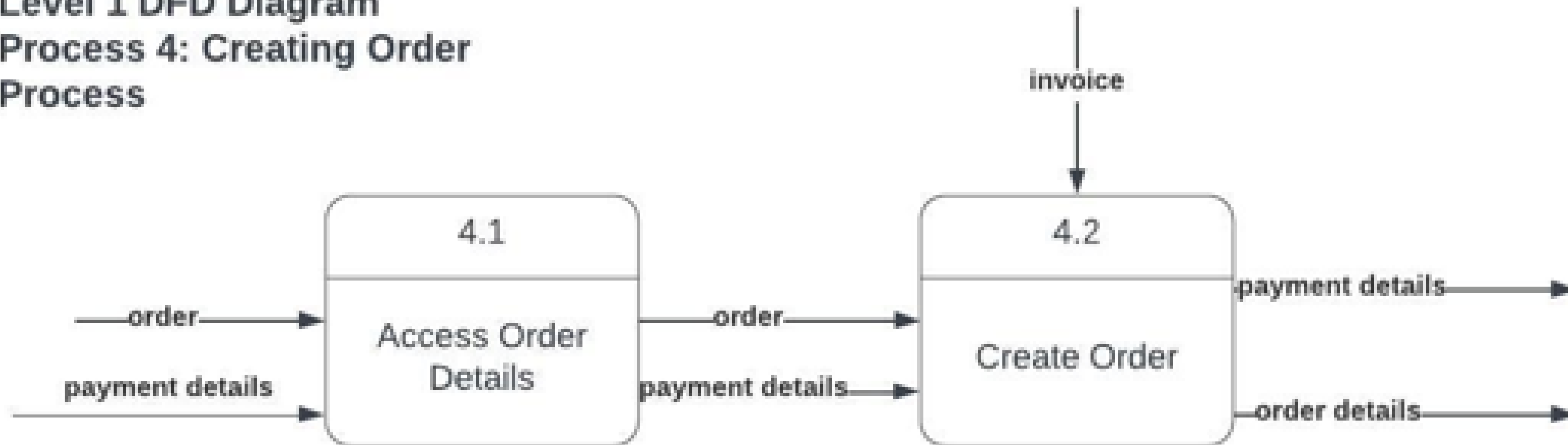
Level 1 DFD Diagram
Process 3: Payment
Process



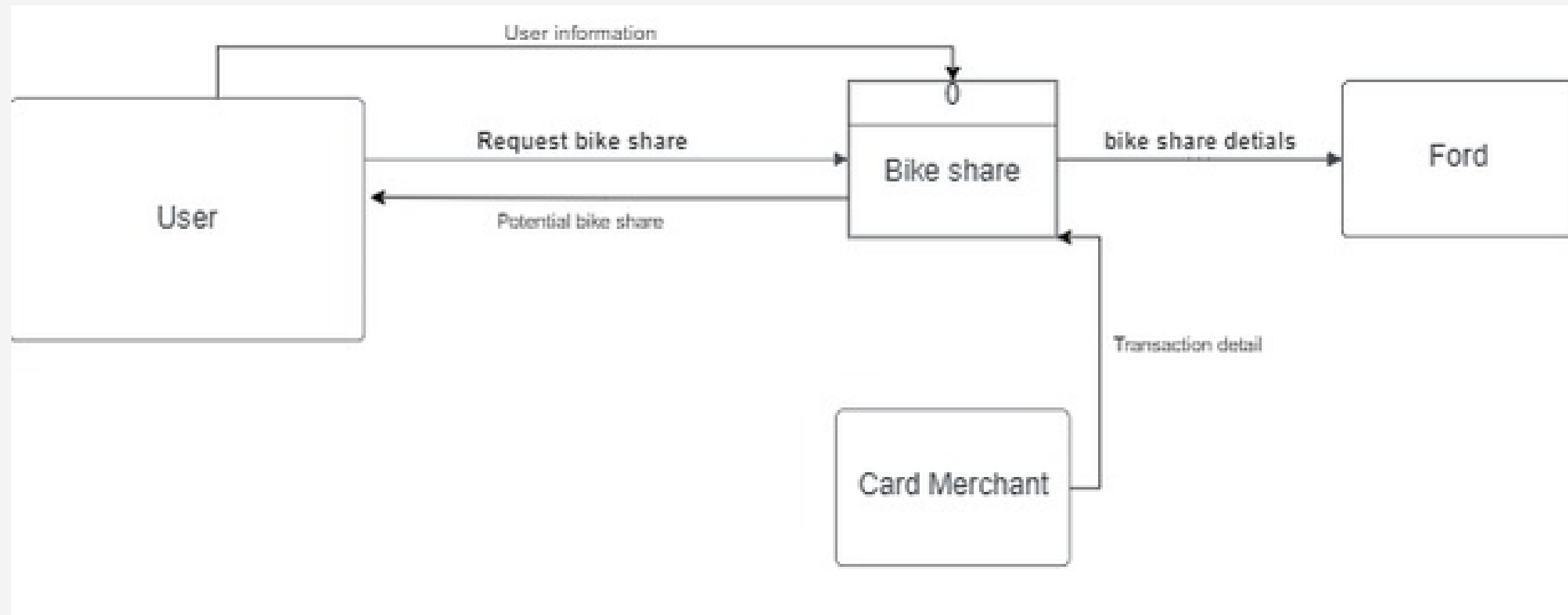
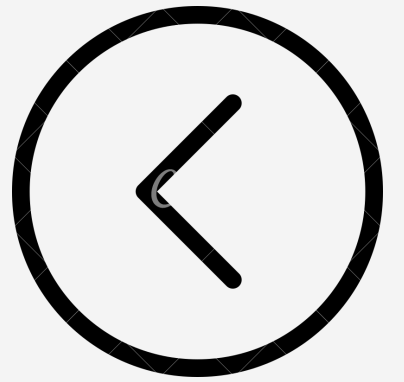
Level 1 DFD Loyalty Program



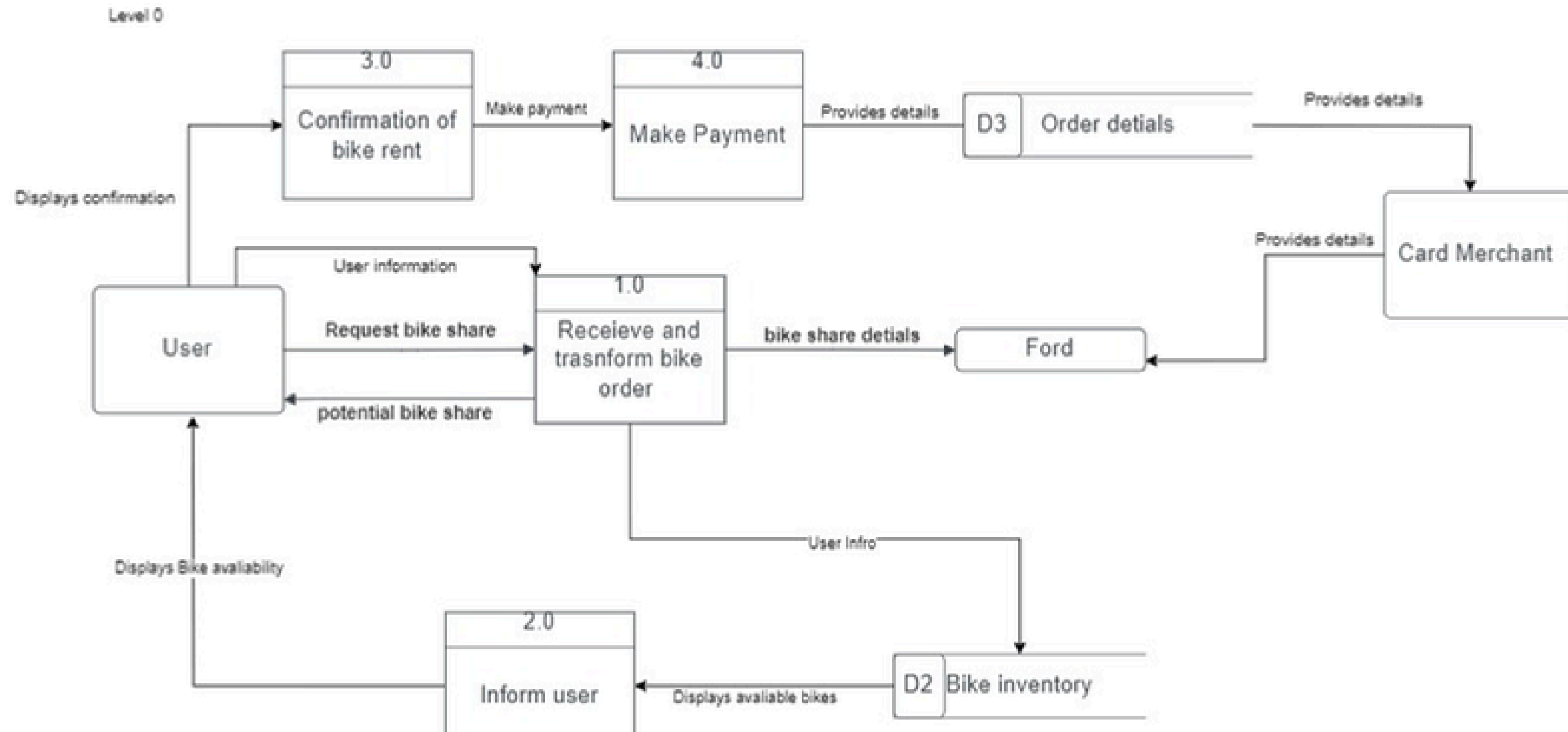
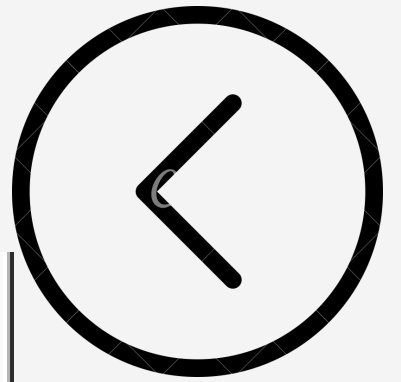
Level 1 DFD Diagram
Process 4: Creating Order
Process



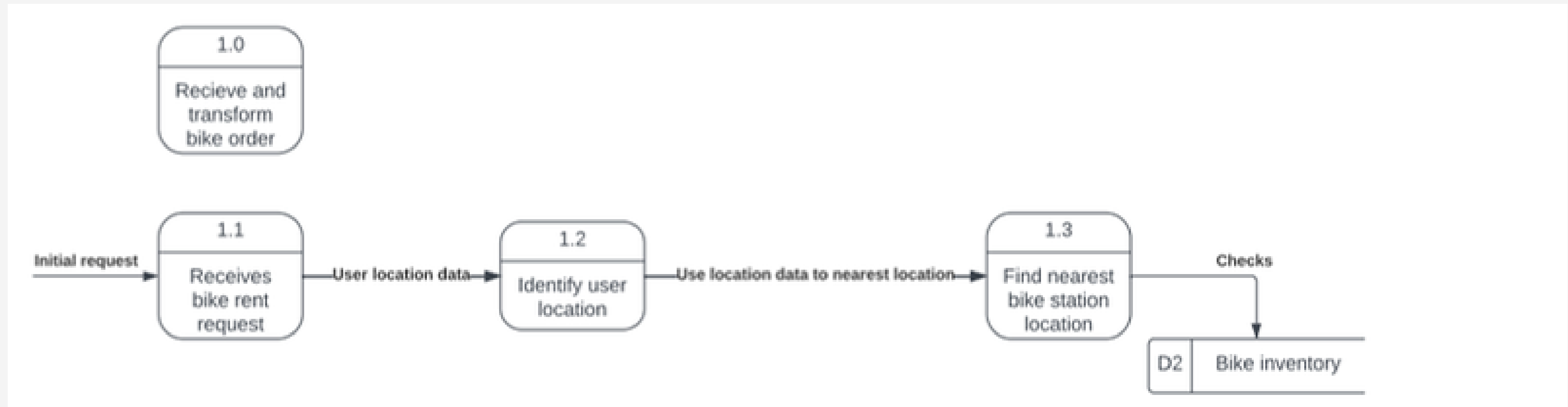
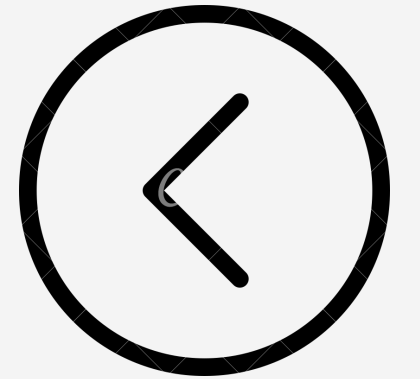
Context Diagram Loyalty Program



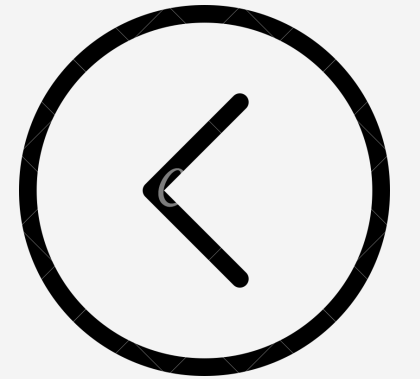
Level 0 Diagram Loyalty Program



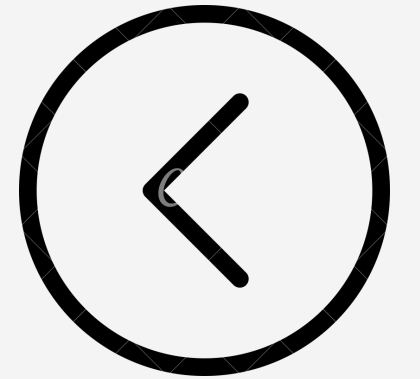
Level 1 DFD Bike share



Level 1 DFD Bike share



Level 1 DFD Bike share



Level 1 DFD Bike share

